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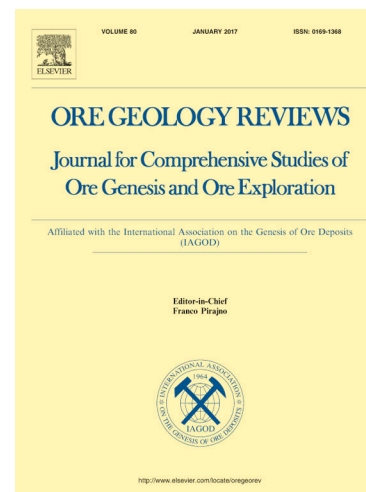
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Critical metals Ga and Ge enrichment in the last-stage and low-temperature sphalerite from the Guojiagou Zn-Pb deposit, western Qinling Orogen, NW China

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20 **Abstract**

21 Gallium (Ga) and germanium (Ge) have been designated as globally critical metals,
 22 driven by their elevating demand and supply risks in high-tech industries such as solar
 23 energy, optoelectronics, and green energy. The Xicheng-Fengtai Orefield, situated
 24 within the Qinling Orogen of Northwest China, contains over 20 million tons of Zn-Pb
 25 metals and thereby is a world-class Zn-Pb orefield hosted in sedimentary rocks.
 26 Nevertheless, the potential for Ga and Ge resources in this region has yet to be
 27 comprehensively assessed. To investigate this matter further, we selected the
 28 representative Guojiagou Zn-Pb deposit to conduct an extensive series of field
 29 investigations complemented by micropetrographic observations and advanced
 30 analytical techniques including electron probe microanalysis (EPMA) and laser
 31 ablation inductively coupled plasma mass spectrometry (LA-ICP-MS). At the
 32 Guojiagou, sulfide mineralization precipitated from brine fluids transporting metals
 33 from the basement, resulting in over 3 Mt of Zn-Pb (with 3.34% Zn and 1.26% Pb), and
 34 was accompanied by extensive ankeritization, silicification, and calcitization. The
 35 multifaceted approach enabled us to elucidate mineral paragenesis, sphalerite
 36 generations, and their chemical compositions. Our analysis revealed three distinct types
 37 of sphalerite: 1) the earliest dark-brown Sp1 intergrown with ankerite; 2) the subsequent
 38 brownish-red Sp2 associated with massive pyrite; and 3) the later light-colored Sp3
 39 linked with calcite veins. EPMA and LA-ICP-MS analyses demonstrated that the last
 40 crystallized Sp3 exhibits significantly elevated concentrations of Ga (mean=139.2 ppm)
 41 and Ge (mean=160.9 ppm), surpassing those found in earlier stages: Sp1 (Ga
 42 mean=20.0 ppm; Ge mean=1.4 ppm) and Sp2 (Ga mean=31.2 ppm; Ge mean=4.6 ppm).
 43 Binary correlation diagrams for specific elements indicate that Ga integrates into
 44 sphalerite's crystal lattice via $\text{Ga}^{3+} + \text{Cu}^+ \leftrightarrow 2\text{Zn}^{2+}$ while Ge incorporates into its
 45 structure through mechanisms such as $\text{Ge}^{4+} + 2\text{Ag}^+ \leftrightarrow 3\text{Zn}^{2+}$ or $\text{Ge}^{2+} \leftrightarrow \text{Cd}^{2+}$ pathways.
 46 Furthermore, GGIMFis sphalerite geothermometer results suggest a decreasing trend
 47 in ore-forming temperatures from approximately 180-230 °C for earlier forms like Sp1
 48 and Sp2 down to about 120-150 °C for late-stage Sp3. Concurrently, sulfur fugacity
 49 values $\log f_{\text{S}_2}$ exhibit a gradual decline from -15 to -10 for both early stages towards -
 50 22 to -14 for late-stage Sp3. The widespread occurrence of ankerite combined with low
 51 manganese concentrations, mainly ranging from 0 to 20 ppm, supports our assertion
 52 that the Guojiagou's ore fluids were relatively oxidizing—these oxidizing fluids mixing
 53 with reduced sulfur generated through thermal reduction of sulfate minerals (TSR),
 54 ultimately leading to substantial precipitation of Zn-Pb ores across the site. Our novel
 55 findings regarding Ga and Ge concentrations—coupled with reserve estimates—
 56 suggest approximate reserves amounting to around 240 tons of Ga alongside
 57 approximately 219 tons of Ge within the Guojiagou Zn-Pb deposit. These discoveries
 58 underscore significant potential and highlight strategic exploration avenues concerning
 59 both Ga and Ge resources present at the Guojiagou and analogous sites throughout
 60 western Qinling Orogen in Northwest China.

61

62 **Keywords:** South Qinling Block; Guojiagou deposit; Zn-Pb deposit; Sphalerite;
63 Gallium; Germanium;

64

65 1. Introduction

66 Gallium (Ga) and germanium (Ge) are critical metals with distinctive properties
67 and extensive applications in green and advanced technologies, including solar energy,
68 optoelectronics, semiconductors, and sustainable energy solutions (Zhao et al., 2017a;
69 Keller & Anderson, 2018; Zheng et al., 2023). Both elements are trace constituents in
70 nature, primarily derived from the continental crust, and are extracted mainly from Zn-
71 Pb sulfides, bauxite, and coal deposits (Cook et al., 2009; Belissont et al., 2014; Frenzel
72 et al., 2016; Schulz, 2017; Wen et al., 2020; Zheng et al., 2023). Nevertheless, the
73 efficient exploration and utilization of Ga and Ge resources remain challenging globally.
74 Recently, over 800 tons of Ga metal reserves have been identified within the tailings of
75 the Fankou MVT Zn-Pb deposit (Yu et al., 2024), positioning similar deposits as prime
76 candidates for future Ga and Ge exploration. However, the mechanisms underlying the
77 enrichment of Ga and Ge in Zn-Pb deposits remain inadequately understood. For
78 instance, some studies suggest Ga concentrates in light-colored sphalerite at lower
79 temperatures (Wang et al., 2023; Wu et al., 2024), while others assert that Ga enriches
80 during high-temperature phases whereas Ge accumulates during low-temperature
81 stages (Moller, 1987). Additionally, some propose that Ge is enriched in earlier-
82 crystallized dark-colored sphalerite while asserting that Ga can accumulate at any stage
83 of mineralization without a distinct enrichment pattern (Meng et al., 2024). Therefore,
84 further case studies are imperative to elucidate the occurrence state, enrichment patterns,
85 and controlling factors influencing Ga and Ge within Zn-Pb deposits.

86 The Xicheng-Fengtai Orefield is situated in the eastern segment of the western
87 Qinling Orogenic Belt (QOB) in northwest China (Figs. 1a and b), encompassing
88 numerous Zn-Pb deposits with reserves exceeding 20 Mt (Yu et al., 2008; Zhang et al.,
89 2015; Wang et al., 2021a). This orefield has experienced significant geological events,
90 including the northward drift of the South Qinling and subsequent subduction-collision
91 between the South China Block (SCB) and North China Block (NCB) (Zhang et al.,
92 2004; Lai et al., 2010; Zheng et al., 2021). Large-scale magmatism during the
93 Indosinian orogeny has modified the original characteristics of orebodies, igniting
94 considerable debate regarding their genesis. A notable consensus exists regarding the
95 origin of these deposits: (1) Orebodies occur in stratiform layers within the Devonian
96 siltstone-carbonate succession; (2) Wallrock alterations are predominantly
97 characterized by silicification and carbonization; (3) The temperature of ore-forming
98 fluids spans a broad spectrum ranging from 380 to 110 °C (Wang et al., 2011a; Ni et al.,
99 2019; Hu et al., 2020); and (4) Ore ages vary from syngenetic Devonian to epigenetic

Late Permian-Triassic (Ma et al., 2004; Hu et al., 2015). Some researchers classify these deposits as SEDEX deposits, asserting that mineralization occurred concurrently with host rock formation (Ma et al., 2004; Ma et al., 2007; Guo et al., 2023), while others advocate for a formation linked to regional magmatic activity (Zhang et al., 2011; Hu et al., 2015; Hu et al., 2024). These two deposit types exhibit distinct differences in host rock relationships, structural features, alteration processes, and ore-forming temperatures, alongside notable variations in trace elements such as Ga, Ge, and In present within sphalerite (Cook et al., 2009; Frenzel et al., 2016). Furthermore, despite the substantial zinc and lead resources in the Xicheng-Fengtai field, there has been scarce research into the substitution mechanisms, storage states, and enrichment patterns of critical metals such as Ga, Ge, and In within sphalerite. Therefore, it is essential to investigate these issues further from the perspective of trace elements hosted in sphalerite, which constitutes the primary objective of our study.

In this study, we conducted a comprehensive geological investigation and sampling in the Guojiagou Zn-Pb deposit, integrating microscopic analysis to elucidate the mineralization processes while systematically selecting various types of sphalerite from different stages for detailed chemical analysis. Investigating the chemical behavior of elements like Ga and Ge within sphalerite across various mineralization stages is essential for a nuanced understanding of metallogenic models. Furthermore, an exhaustive study on the enrichment patterns of Ga and Ge in sphalerite from Zn-Pb deposits within this region can provide valuable insights and strategic guidance for prospecting these elements throughout the QOB.

2. Region geology

The western QOB constitutes the western segment of the east-west-trending QOB, delineating the boundary between the North China Craton and the Yangtze Craton. The QOB has undergone a protracted and complex evolutionary history, with its earliest origins tracing back to the Neoproterozoic-Paleoproterozoic era when the crystalline basement was formed (Zhang et al., 2023; Yuan et al., 2022). During the Devonian to Carboniferous periods, multiple transgressions and regressions resulted in extensive deposition of marine sedimentary sequences. Since the Triassic, the collision between the North China Craton and the Yangtze Craton has led to the formation of QOB, subsequently followed by intracontinental orogeny characterized by widespread magmatism (Li et al., 2007; Dong & Santosh, 2016; Zhao et al., 2017b; Wang et al., 2024a). These intense tectonic events, along with associated sedimentation and magmatism, have significantly contributed to a rich endowment of metal deposits such as Zn-Pb, Au, Ag, and Mo (Mao et al., 2002; Li et al., 2012; Leach & Song, 2019; Qiu et al., 2021; Yu et al., 2022a; Qiu et al., 2024).

The Xicheng-Fengtai Orefield, located in the eastern segment of the western QOB, serves as one of the most significant Zn-Pb metal producers both in China and

worldwide, hosting approximately 80 Zn-Pb deposits and prospects with estimated metal resources exceeding 20 Mt (Yu et al., 2008; Zhang et al., 2015; Wang et al., 2021a). A series of east-west-trending faults and folds have been extensively developed within the Xicheng-Fengtai Orefield as a result of the collision between the North China Craton and the Yangtze Craton. Among these structures, the Fengxian-Taibai-Shixia (FTS) Fault plays a crucial role in controlling the distribution of major strata, intrusions, and ore deposits (Fig. 1b), being interpreted as a primary conduit for ore fluid migration (Ma et al., 2004).

A significant thickness of Paleozoic strata is extensively exposed in the western QOB, characterized by lithological compositions ranging from shallow marine to littoral facies limestone and phyllite. The Devonian sequences, notable for their considerable thickness, unconformably overlie the Silurian sequences and are unconformably overlain by the Carboniferous sequences. Furthermore, the Devonian strata are subdivided into four formations—Wujiashan, Anjiachagou, Xihanshui, and Dongshan—in ascending order (Fig. 1c). The Wujiashan Formation is lithologically composed of schist and marble, indicative of relatively intense metamorphism. In contrast, the overlying Middle to Upper Devonian sequence consists of weakly silicified limestone and phyllite. Above these layers, the Carboniferous to Triassic strata are characterized by unmetamorphosed or weakly metamorphosed biogenic limestones and clastic sedimentary rocks.

3. Deposit geology

The Guojiagou Zn-Pb deposit is categorized into two distinct ore belts: the northern and southern ore belts, encompassing a total of 80 orebodies, with 52 located within the northern ore belt (Fig. 2).

3.1 Host rocks

At the Guojiagou deposit, the Middle Devonian Anjiacha Formation is subdivided into two members: the Changba Member and the Jiaogou Member. The silicified limestone of the Jiaogou Member serves as the primary ore-hosting rock, with most Zn-Pb mineralization occurring predominantly along the interface where limestone contacts phyllite. Subsequent structural and magmatic activities have profoundly influenced the distribution of orebodies, resulting in steeper angles of strata and orebodies near the Mishuling pluton (Pu, 2018; Zhang et al., 2019b).

3.2 Ore-controlling structure

Both faults and associated folds are widespread at the Guojiagou, significantly controlling the distribution of major Zn-Pb orebodies (Fig. 2). Three sets of faults have been identified: (1) NWW-NEE to nearly EW-trending, (2) NNW-SSE to SSW-trending, and (3) NE-trending faults. Among these, the NWW-trending F2 and F5 faults,

subordinate to the regional FTS Fault, control the distribution of strata, orebodies, and magmatic plutons (Zhang et al., 2019a). F2 and F5 are north-dipping thrust faults with dip angles between 50° and 80° and widths of tens of meters. The eastern section of the Luoba Anticline serves as the primary location for orebodies; its axis trends EW and dips eastward at angles from 2° to 8° (Zhang, 2019). Additionally, numerous secondary anticlines and synclines are well-developed and directly regulate most Zn-Pb orebodies. Notably, orebodies in the southern belt exhibit greater thicknesses and higher grades than those in the northern belt (Pu, 2018).

3.3 Orebodies geometry and distribution

The north and south ore belts together form an east-west elongating zone approximately 5 km in length and 2 km in width (Fig. 2). All orebodies are characterized as stratiform and stratabound, with significant burial depths exceeding 300 m (Fig. 3). Three distinct types of orebodies have been identified based on their occurrences and host rocks: (1) orebodies located within lenticular limestone interbedded with phyllite, (2) orebodies situated along the limestone side of the contact zone between phyllite and limestone, and (3) orebodies found within a thick limestone layer at the saddle of the anticline. Notably, the second type occurring at the limestone-phyllite interface represents the predominant occurrence of these orebodies.

Compared to the southern ore belt, the orebodies in the northern ore belt are characterized by relatively shallow burial and lower grades (Fig. 3a, Pu, 2018). The occurrences of these orebodies in the northern ore belt exhibit a gentle eastward dip with angles less than 5°. In contrast, the orebodies in the southern ore belt display a steeper inclination toward the north, with angles ranging from 20° to 40° (Fig. 3b), taking on a U-shaped configuration in vertical section view. Although there is variability in the occurrences of orebodies within both belts, they generally align with those of their respective host strata.

3.4 Feature of alterations

Ankeritization, silicification, and calcitization represent the primary alterations spatially, temporally, and genetically associated with Zn-Pb mineralization at the Guojiagou deposit (Figs. 4a-c), controlled by fault systems and specific lithological layers. Ankeritization is extensively developed throughout the entire Guojiagou Zn-Pb deposits, constituting an east-west linear alteration zone that is present throughout the ore-forming process. Most Zn-Pb orebodies are hosted within silicified limestone, where ore grades and thickness significantly decrease into bioclastic limestone; thus, silicification is also closely linked to major Zn-Pb mineralization at the Guojiagou. Comparably, calcitization primarily occurs in the mid to late stages of mineralization, often accompanied by calcite veins and crystal cavity structures locally. Moreover, a minor presence of organic matter, including bitumen and graphite, is predominantly observed within the host rocks, where sulfides are occasionally penetrated or intricately

215 intergrown with them (Fig. 4d).

216 3.5 Paragenetic sequence

217 At the Guojiagou, sphalerite, galena, pyrite, and iron sulfides (including marcasite
218 and pyrite) are the predominant minerals, accompanied by minor occurrences of
219 chalcopyrite, arsenopyrite, and tetrahedrite. The gangue minerals include ankerite,
220 calcite, dolomite, quartz, and minor siderite, in decreasing order of abundance. Based
221 on mineral assemblages as well as cross-cutting and replacement relationships observed
222 in the samples, we identified three distinct ore periods: (1) the pre-ore period, (2) the
223 main mineralization period, and (3) the post-ore period. Furthermore, the main ore
224 period is further subdivided into three distinct stages: (1) stage 1 (characterized by
225 sedimentation and diagenesis), (2) stage 2 (marked by infilling and replacement
226 processes), and (3) stage 3 (involving reworking) (Fig. 6).

227 3.5.1 Pre-ore period

228 This sedimentary and diagenetic stage is synchronous with the formation of host
229 rocks, accompanied by the deposition of fine-grained ankerite (Ank0) and calcite (Cc0)
230 (Fig. 5a). Framboidal pyrite (1 – 10 μm in diameter), typically observed as isolated
231 grains or small clusters, is conspicuously less abundant in ore-bearing horizons, likely
232 reflecting its recrystallization and aggregation under the influence of metasomatic
233 processes driven by metal-bearing hydrothermal fluids. During this diagenetic process,
234 sphalerite and galena were observed to replace pyrite, resulting in the formation of
235 framboidal pseudomorphs of sphalerite and galena (Fig. 5b). The presence of organic
236 matter within the host rocks, subsequently replaced by sphalerite and galena during the
237 main-ore period, likely reflects its derivation from adjacent host rocks and its transport,
238 concentration, and metamorphic transformation through brine migration (Fig. 4d).

239 3.5.2 Main-ore period

240 Stage 1

241 Following extensive silicification and ankeritization in a later stage, a complex
242 assemblage of red-brown vein-like and massive ankerite (Ank1), ranging from tens of
243 centimeters to meters in scale, was formed and intruded into the interlayers of the host
244 rocks (Fig. 5c). Additionally, minor fine-grained siderite was observed among coarse
245 ankerite crystals, along with precipitation of calcite veins (Cc1) and various sulfides
246 that constituted a dull red halo at the contact between ankerite layers (Fig. 5c).
247 Subsequently, the black-brown sphalerite (Sp1) emerged, predominantly exhibiting a
248 fine-grained colloform texture while revealing opaque characteristics under transmitted
249 light; it is notably characterized by extensive “chalcopyrite disease” (Figs. 4b, 5c-d).
250 Meanwhile, framboidal pyrite gradually aggregated to form medium-coarse subhedral-
251 euhedral textures; however, galena content remained relatively low and exhibited

subhedral-anhedral textures within coarse-grained veinlets. In addition, during this stage, minor bitumen is observed in irregular speckled forms coexisting with sphalerite.

Stage 2

The continuous replacement of ankerite and calcite has led to the development of massive dissolution features, including collapse structures, cavities, and breccias (Fig. 5e). A significant amount of Zn-Pb sulfides was deposited in conjunction with these characteristic infilling and replacement ore textures. Additionally, it is evident that variable sulfides and layered structures of ankerite form around the periphery of the crystal cavities (Figs. 4b and 5f). Subsequently, a substantial quantity of coarse- to fine-grained pyrite and sphalerite infiltrated and replaced the ankerite (Fig. 5f). At this stage, a significant amount of pyrite emerges, exhibiting continuous growth accompanied by a gradual increase in grain, with remnants of framboidal pyrite occasionally preserved at the cores of euhedral crystals (Fig. 5g). Meanwhile, sphalerite exhibits a paler hue, appearing dark red in transmitted light. In comparison, fine-grained sphalerite partially enclosed by pyrite appears colorless (Figs. 5g and h). Galena precipitates later in this sequence and is closely associated with dolomite (Fig. 5i).

Stage 3

The defining characteristic of this stage is the presence of numerous calcite-sulfide veins, with a predominant mineral assemblage comprising coarse-grained calcite (Cc3), ankerite (Ank3), and Zn-Pb sulfides in decreasing order of abundance. Sphalerite exhibits a lighter hue and enhanced transparency, ranging from light reddish-brown to yellowish-brown, and occurs within pure calcite veins (Figs. 5k and l). Additionally, aggregates of sphalerite can be observed within the host rocks outside the fractures of coarse-grained sphalerite-calcite veins (Fig. 5j).

3.5.3 Post-ore period

Later, coarse-grained vein-like calcite (Cc4) traverses all previous mineral assemblages. It appears relatively pure, predominantly displaying a pristine white color while occasionally interspersed with minor occurrences of delicate veinlets (Fig. 5j). During this period, the sulfide content is extremely low, predominantly represented by micro-fine grained, colorless, and transparent sphalerite under the transmitted light microscope. It occurs as speckled distributions along both sides of calcite veins, inferred to have formed through the activation, modification, and subsequent reprecipitation of earlier sphalerite by the calcite veins.

4. Samples and analytical methods

4.1 Sampling

In this study, a total of 96 samples were collected from drill cores and underground

workings, encompassing diverse Zn-Pb ores, host rocks, and granite from both the southern and northern ore belts. We systematically categorized all samples and conducted detailed petrographic examinations to elucidate the sequences of mineral formation. Subsequently, we selected representative sphalerite specimens exhibiting various mineral assemblages and distinct mineralogical characteristics for further electron probe microanalysis (EPMA) and laser ablation inductively coupled plasma mass spectrometry (LA-ICP-MS) measurement. By performing EPMA and LA-ICP-MS sequentially on identical points, we ensured that our results were comparable, enhancing our analyses' accuracy and reliability.

4.2 EPMA analysis

Quantitative compositional analysis of sphalerite was performed using a JEOL JXA-iSP100 electron probe microanalyzer at the Southern Marine Science and Engineering Guangdong Laboratory (Zhuhai, China). The instrument operated under specific conditions, including an accelerating voltage of 20 kV, a probe current of 20 nA, and a beam diameter of 5 μm . The X-ray lines and standards employed were as follows: S K α , Zn K α , Fe K α , Mn K α , Cd L α , Ga K α , Ge K α , In L α , Cu K α , and Sn L α . Peak counting times were set to 10 seconds for major elements such as S, Zn, and Fe while extended to 20-50 seconds for minor elements including Mn, Cd, Ga, Ge, In, Cu and Sn. Background counting times were established at half the peak durations for both high- and low-energy background positions. The ZAF correction method was utilized to ensure result accuracy by accounting for factors such as atomic number effects along with absorption and fluorescence corrections. Calibration standards used in the analysis included sphalerite for Zn; pyrite for Fe and S; manganese oxide for Mn; greenockite for Cd; gallium phosphide for Ga; germanium dioxide for Ge; indium phosphide for In; chalcopyrite for Cu; and tin metal for Sn. The minimum detection limits in sphalerite were determined as follows: S (54 ppm), Zn (222 ppm), Fe (82 ppm), Mn (77 ppm), Cd (91 ppm), Ga (148 ppm), Ge (155 ppm), In (93 ppm), Cu (112 ppm), and Sn (93 ppm).

4.3 In situ LA-ICP-MS trace elements analysis (spot)

Trace elements analysis of sphalerite was conducted using a New Wave Research (NWR) 193 nm ArF Excimer laser ablation system, paired with a Thermo Scientific iCAP RQ quadrupole inductively coupled plasma mass spectrometer (ICP-MS) at Guangzhou Tuoyan Analytical Technology Co., Ltd., in Guangzhou, China. The laser ablation system was calibrated using the NIST SRM610 standard glass to optimize low oxide production rates. During the analyses, we utilized a helium gas flow of 700 ml/min as the carrier gas, which combined with an argon gas flow of 0.89 L/min to fine-tune the system's sensitivity. The laser operated at a fluence of 3.5 J/cm², with a repetition rate of 6 Hz and a spot diameter of 30 μm . Each analysis included a 45-second period for laser ablation followed by a 40-second background measurement. We analyzed various trace elements such as ²⁴Mg, ³⁴S, ⁵⁵Mn, ⁵⁷Fe, ⁵⁹Co, ⁶⁰Ni, ⁶⁵Cu, ⁶⁶Zn,

⁷¹Ga, ⁷²Ge, ⁷⁵As, ⁷⁷Se, ¹⁰⁷Ag, ¹¹¹Cd, ¹¹⁵In, ¹¹⁸Sn, ¹²¹Sb, ¹⁹⁷Au, ²⁰²Hg, ²⁰⁵Tl, ²⁰⁸Pb and ²⁰⁹Bi. The concentrations were calibrated using external standards, specifically NIST SRM610 and MASS-1 reference materials. To ensure data quality, we used sulfide standards SRM612 and GE8 as unknown samples (Wilson et al., 2002). The raw data from the laser ablation process were processed offline using the IOLITE software package's 3D Trace Elements data reduction scheme (Paton et al., 2011). This processing involved corrections for instrument sensitivity drift, baseline subtraction, and calculations for trace elements concentrations. For detailed procedures regarding LA-ICP-MS analysis and data processing methods, please refer to Liu et al. (2008) and Chu et al. (2023).

4.4 Principal component analysis of trace elements data

To further explore the relationship between various trace elements and samples, we conducted a principal component analysis (PCA) for dimensionality reduction. The underlying principle is as follows: by performing an orthogonal transformation, PCA converts a set of potentially correlated variables into a set of linearly uncorrelated variables while preserving as much original information as possible. These new variables are ordered according to their variance, with the first principal component (PC1) capturing the greatest variance and thereby reflecting the most significant relationships among the variables. If PC1 alone is insufficient to represent the information contained in the original data, additional components—such as the second principal component (PC2)—may be considered. To ensure reliable analysis, we established specific criteria, including: (1) element concentrations should exceed detection limits in over 70% of cases; (2) influences from micro-inclusions should be minimized; and (3) values below detection limits should be imputed to maintain dataset integrity.

5.Result

5.1 EPMA data of sphalerite

A total of 90 EPMA points were analyzed, comprising 25 from Sp1, 31 from Sp2, and 34 from Sp3. The detailed results are presented in Table 1. Sulfur and zinc, as the primary elements in sphalerite, exhibit minimal variation across the three types of sphalerite, with comparable concentrations. For instance, in Sp1, sulfur ranges from 32.1% to 33.0% and zinc from 62.2% to 64.9%, while in Sp2 sulfur ranges from 32.0% to 32.9% and zinc spans from 61.0% to 63.8%. In Sp3, sulfur varies between 31.8% and 32.8%, while zinc fluctuates between 60.9% and 64.0%. Differently, iron (Fe), the most significant minor element in sphalerite, displays a broader range of variation ranging between 1.1% and 5.1%. Cadmium also exhibits notable variability as another important minor element with values ranging from 0.24% to 0.48%. However, Ga and Ge—elements of particular interest—are generally below detection limits; only two points in Sp3 exceed measurable limits.

366 5.2 LA-ICP-MS data of sphalerite

367 We obtained 90 LA-ICP-MS analyses corresponding to the EPMA examination,
 368 including 25 from Sp1, 31 from Sp2, and 34 from Sp3. The trace elements
 369 concentrations in sphalerite and their statistical ranges, means, and medians are detailed
 370 in Table 2 and Fig 7. Overall, the trace elements exhibit varying degrees of enrichment;
 371 Fe, Cd, Cu, Ge, and Ga show the relatively high concentrations, followed by Pb, Ag,
 372 Sb, Sn, Hg, and Mn. In contrast, Bi, Ni, Co and In are mostly below detection limits
 373 (BDL). Fig 8 presents laser ablation profiles for three representative sphalerite samples
 374 to further illustrate the occurrence and variation of trace elements within sphalerite.
 375 Notably, the concentration of trace elements within a single sphalerite grain can vary
 376 significantly by up to two to three orders of magnitude.

377 5.2.1 Sp1 compositions

378 A limited number of elemental signals can be detected in the laser ablation analyses,
 379 including Zn, S, Fe, Cd, Hg, Ga, and Pb (Fig. 8a). These elements exhibit signal lines
 380 parallel to the Zn signal with a relatively flat pattern, suggesting lattice substitution
 381 within sphalerite structure. In contrast, elements such as Cu, Ag, Mn, Sb, Sn, and Ge
 382 display lower and more fluctuating signals, indicative of the presence of solid solution
 383 or nano-mineral inclusions. The detectable trace elements include Mn, Fe, Co, Cu, Ga,
 384 and Ge. Manganese concentrations are relatively low at 3.1 to 44.2 ppm with three
 385 outliers at 171, 318, and 479 ppm. Iron concentrations vary significantly from 10503 to
 386 29260 ppm while Cadmium concentrations are more narrowly distributed between
 387 2529 to 4079 ppm. Gallium concentrations show considerable variability, ranging from
 388 0.5 to 108.5 ppm with an average of 20.0 ppm; meanwhile, germanium concentrations
 389 are relatively low, ranging from 0.5 to 3.1 ppm with an average of 1.4 ppm.

390 5.2.2 Sp2 compositions

391 Only a spectrum of elements, including Zn, S, Fe, Cd, and Hg, are detectable in the
 392 laser ablation signals (Fig. 8b). These elements exhibit signal lines that run parallel to
 393 the Zn signal and display relatively flat profiles, also suggesting lattice substitution
 394 within sphalerite structure. In contrast, elements such as Ga, Pb, Cu, Ag, Mn, Sb, Sn,
 395 and Ge show relatively lower and more variable signals likely indicative of their
 396 presence in solid solutions or as nano-mineral inclusions. Among the detected trace
 397 elements, manganese stands out with higher concentrations than Sp1, ranging from 5.4
 398 to 44.5 ppm and including an outlier at 318.1 ppm. Iron concentrations are generally
 399 elevated relative to those in Sp1, varying from 15179 to 33732 ppm. Cadmium
 400 concentrations are both higher and more narrowly distributed than those in Sp1, ranging
 401 from 2848 to 3963 ppm. Gallium levels are comparable to those in Sp1, spanning from
 402 1.9 to 80.0 ppm with an average of 31.2 ppm; meanwhile, germanium concentrations
 403 exhibit greater uniformity, ranging from 0.4 to 65.8 ppm with an average of 4.6 ppm.

5.2.3 Sp3 compositions

In Sp3, a less range of elements—including Zn, S, Fe, Cd, Hg, Cu, and Ge—have been detected (Fig. 8). These elements exhibit signal lines parallel to the Zn signal with relatively flat profiles, indicating lattice substitution within sphalerite structure. In contrast, elements such as Ga, Pb, Ag, Mn, Sb and Sn display lower and more fluctuating signals that suggest their presence in solid solutions or as nano-mineral inclusions. Among the detected trace elements, manganese concentrations range from 3.6 to 27.6 ppm with an average of 12.0 ppm; notably, no anomalous concentrations are observed unlike in Sp1 and Sp2. Iron concentrations are generally comparable to those in Sp2, ranging from 11540 to 33991 ppm with an average of 25155 ppm. Cadmium concentrations align closely with those in Sp1 and Sp2, narrowly distributed between 2523 and 3902 ppm. It is noteworthy that gallium is most enriched in Sp3 compared to Sp1 and Sp2, with concentrations ranging from 21.7 to 514.1 ppm and averaging 139.2 ppm. Germanium also exhibits significant enrichment with great variation, ranging from 1.4 to 621.5 ppm, with an average of 160.9 ppm.

6. Discussion

6.1 Occurrences of Ga and Ge in the Guojiagou sphalerite

Sphalerite is widely recognized for its ability to accommodate a diverse array of trace elements, including Fe, Mn, Ga, Ge, and In. Three primary mechanisms of incorporation have been identified: (1) individual micro-mineral inclusions, (2) direct stoichiometric substitution, and (3) coupled stoichiometric substitution within the crystal lattice (Cook et al., 2009; Ye et al., 2011; Frenzel et al., 2016; Cugerone et al., 2018). Individual micro-mineral inclusions can be directly observed using high-resolution optical or electron microscopy. Divalent cations such as Fe^{2+} , Mn^{2+} , and Cd^{2+} —which possess ionic radii comparable to that of Zn^{2+} —can substitute directly into the sphalerite lattice at levels reaching several tens of weight percent (Liu et al., 2010; Bonnet et al., 2016; Shu et al., 2024; Chen et al., 2024a). In contrast, for higher-valent ions such as Ga^{3+} , Ge^{4+} , and In^{3+} substituting for Zn^{2+} , the incorporation process occurs through a coupled stoichiometric substitution mechanism that involves both high-valent and low-valent ions like Cu^+ and Ag^+ to achieve charge balance and ionic radius similarity (Cook et al., 2009; Ye et al., 2011; Murakami & Ishihara, 2013; Wang et al., 2024b).

In the Guojiagou deposit, our micropetrographic observations reveal the presence of chalcopyrite, tetrahedrite, and pyrite micro-mineral inclusions within Sp1 and Sp2 (Figs. 5d, f, g, and h). Consequently, a portion of Cu and Fe is inferred to exist as micro-mineral inclusions in sphalerite. In the meantime, we have converted the elemental content units from ppm to $\mu\text{mol/g}$ to establish a more precise quantity relationship between the elements.

Incorporating the LA-ICP-MS signal evidence, the data presented in Fig. 9a demonstrate a significant negative correlation between the concentrations of elements such as Fe, Mn, and Cd with Zn. This supports the hypothesis that these elements, possessing ionic radii comparable to that of Zn^{2+} , can effectively substitute for Zn within the sphalerite lattice. Meanwhile, the distribution of Cd and Fe shows a positive correlation for Sp3 but no strong correlation for Sp1 and Sp2, suggesting that Cd and Fe may couple to substitute Zn in Sp3 (Liu et al., 2010). Consequently, Fe, Mn, and Cd are incorporated into the sphalerite lattice through direct substitution mechanisms represented as $\text{Fe}^{2+} \leftrightarrow \text{Zn}^{2+}$, $\text{Mn}^{2+} \leftrightarrow \text{Zn}^{2+}$, $\text{Fe}^{2+} + \text{Cd}^{2+} \leftrightarrow 2\text{Zn}^{2+}$, and $\text{Cd}^{2+} \leftrightarrow \text{Zn}^{2+}$.

The Ga vs. Cu binary diagram shows that all data points have a positive linear correlation and are positioned to the left of $(\text{Cu}/\text{Ga})_{\text{mol}} = 1$ (Fig. 9b), indicating that Cu concentrations exceed those of Ga. In contrast, some points fall to the right of the $(\text{Ag}/\text{Ga})_{\text{mol}} = 1:1$ line in the Ga vs Ag binary diagram, suggesting that Ag content with Ga is insufficient for effective Zn substitution. This trend also exhibits a weak negative linear correlation ($R^2 = 0.01$), highlighting the limited association between Ag and Ga in this mechanism (Fig. 9c). The ionic radii of Ga^{3+} (62 Å), Zn^{2+} (74 Å), and Cu^+ (96 Å) are similar; however, Ag^+ has a larger ionic radius at 126 Å, which affects structural stability. This explains the low Ag content in the Guojiagou sphalerite (<10 ppm). We propose that Ga^{3+} enters sphalerite primarily through coupled substitution with Cu^+ , while Ag contributes minimally to Ga enrichment.

No clear correlation was observed between the Ge vs. Cu and Ge vs. Ag plots (Figs. 9d and e). A small subset of points lies to the right of the $(\text{Cu or Ag})/\text{Ge}_{\text{mol}} = 2:1$ line, indicating that higher concentrations of Ge do not couple with Cu or Ag for Zn substitution. In contrast, at lower concentrations, Ge shows a positive linear correlation (slope close to 0.5) with Ag (Fig. 9e). Additionally, when examining relationships with Cd, Mn, Fe, Pb, and other elements, we found that Ge—especially at elevated concentrations—displays a negative linear correlation (slope close to 1) with Cd (Fig. 9f). Principal Component Analysis shows two opposing arrows for Cd and Ga, indicating a negative correlation between these elements (Fig. 10). Thus, we propose that Ge is incorporated into the Guojiagou sphalerite through two mechanisms: at lower concentrations as $\text{Ge}^{4+} + 2\text{Ag}^+ \leftrightarrow 3\text{Zn}^{2+}$; and higher concentrations as $\text{Ge}^{2+} \leftrightarrow \text{Cd}^{2+}$.

We also investigated the relationships between Cu and various trivalent and tetravalent cations, revealing significant trends. Notably, In and Sn show a strong positive linear correlation (Fig. 9g) and a positive relationship with Cu (Fig. 9h), suggesting their coupled substitution for Zn in sphalerite is a key mechanism. Conversely, Ag does not exhibit a similar correlation, indicating that Cu primarily facilitates high-valence cation substitution as the partner ion. Additionally, a strong positive correlation was observed between Sb and Cu (Figs. 9i and 10), further supporting that the substitution mechanism likely follows the equation $2\text{Cu}^+ + \text{Sb}^{4+} \leftrightarrow 3\text{Zn}^{2+}$.

482 6.2 Physicochemical conditions of Zn-Pb mineralization

483 The physicochemical parameters, such as temperature, pressure, solution
 484 compositions, Eh, and pH, significantly influence the concentration of trace elements
 485 within minerals during their formation. Consequently, the concentration and
 486 distribution of trace elements in minerals act as indicators for reconstructing the
 487 physicochemical environment. Sphalerite is particularly instrumental in elucidating the
 488 relationship between its elemental composition and the surrounding physicochemical
 489 conditions (Lockington et al., 2014; Frenzel et al., 2021; Chen et al., 2024b). For
 490 instance, sphalerite crystallizing at elevated temperatures is commonly enriched in Fe,
 491 Mn, and In, whereas that formed at lower temperatures typically exhibits enrichment in
 492 Ga and Ge (Frenzel et al., 2016; Bauer et al., 2019; Wu et al., 2023). Consequently,
 493 several qualitative parameters such as Ga/In, Zn/Fe, Zn/Cd, and log(Ga/Ge) can be
 494 utilized to assess the crystallization temperature (Han, 1994; Liu et al., 1984; Möller,
 495 1987; Yu et al., 1987).

496 In the Guojiagou deposit, the concentrations of trace elements such as Fe, Mn, Ga,
 497 Ge, and In in sphalerite are significantly enriched and typically exhibit considerable
 498 heterogeneity, with variations spanning several orders of magnitude within a single
 499 crystal. We utilized these ratios to qualitatively analyze the metallogenic temperature
 500 of the Guojiagou deposit, obtaining relatively high Ga/In ratios (0.49 - 4165), Zn/Fe
 501 ratios (18.15 - 61.61) and Zn/Cd ratios (157.57 - 252.79) suggesting that sphalerite
 502 formed in a medium- to low-temperature environment (Liu et al., 2010; Ye et al., 2012;
 503 Gao et al., 2016). To gain more precise temperatures, we applied the GGIMFis (Ga, Ge,
 504 In, Mn and Fe in sphalerite) sphalerite geothermometer to calculate the crystallization
 505 temperatures of sphalerite (Indium concentrations below detection were replaced with
 506 0.7 times the limit) (Frenzel et al., 2016). The results indicate that the temperatures of
 507 Sp1 and Sp2 range from 180 to 230°C, while Sp3 crystallized at lower temperatures
 508 ranging from 120 to 150°C (Fig. 11a). These findings align with previous fluid
 509 inclusion thermometry results ranging from 123 to 318°C, peaking at 130 to 230°C
 510 (Wang et al., 2011a; Ni et al., 2019; Hu et al., 2020). Furthermore, our analysis reveals
 511 that both Ga and Ge concentrations gradually increase with decreasing temperature over
 512 time (Figs. 11b and c).

513 In the Guojiagou deposit, Sp1 and Sp2 are associated with substantial pyrite,
 514 whereas Sp3 is linked to minimal pyrite content. The sulfur fugacity (fS_2) for each stage
 515 was calculated using the formula from Frenzel et al. (2021), revealing that the $(FeS)_{mol}$
 516 content in sphalerite ranges from 0.01% to 1.0%, placing it within the intermediate
 517 sulfide range. The Log fS_2 values for Sp1 and Sp2 range from -15 to -10, while those
 518 for Sp3 span -22 to -14 (Fig. 11d). The gradual decrease in fS_2 values from Sp1 and
 519 Sp2 to Sp3 indicates reduced availability of sulfur in the hydrothermal fluid system
 520 during mineralization. Since fS_2 values during pyrite formation are generally higher
 521 than those for sphalerite, the observed decline in pyrite content from early to late stages

further corroborates these intermediate and decreasing fS_2 values (Fig. 5).

Moreover, the Mn content in sphalerite serves as an indicator of redox conditions during crystallization, with elevated concentrations suggesting reducing environments and lower concentrations indicating more oxidizing conditions (Kelley et al., 2004; Maslennikov et al., 2009). In the Guojiagou deposit, Mn levels in sphalerite mainly range from 0 to 20 ppm, implying that mineralization occurred under predominantly oxidizing conditions. Zhang (2019) analyzed various mineral inclusions within the Guojiagou deposit and found that the mineralizing fluid was mildly acidic (pH = 5.60 - 5.65) and oxidizing (Eh = 0.22 - 0.32), which corroborates our conclusions. The deposit also contains abundant ankerite throughout the mineralization process. Cook et al. (2000) noted that Zn-Pb-bearing oxidizing fluids can form siderite under mildly acidic conditions, leading us to hypothesize that the mineralizing fluids were both mildly acidic and oxidizing.

6.3 Ore genesis and critical metals Ga and Ge potential at the Guojiagou

The Zn-Pb deposits in the Xicheng-Fengtai orefield have long been a focal point of debate regarding their genetic classification as SEDEX, magmatic hydrothermal, or MVT deposits (Ma et al., 2004; Hu et al., 2015; Pu, 2018; Zhang et al., 2019a). Tectonically, the Xicheng-Fengtai orefield is situated at the western QOB, where it converged with the North China Craton and the Yangtze Craton during the Triassic period (Dong & Santosh, 2016; Yu et al., 2022b; Qiu et al., 2023). This collision resulted in substantial granite formation (Guo et al., 2011; Li et al., 2013; Wei et al., 2017; Qiu et al., 2020; Yu et al., 2021). Prior to this event, the western QOB was located within a marginal basin of the Yangtze Block and hosted thick Devonian to Carboniferous marine sequences comprising albite, siliceous rocks, carbonates, and ankerite (Ma et al., 2004; Dong et al., 2013; Fang, 2023).

Irish-type deposits are stratabound, depositing soon after host rock, carbonate-hosted sulfide bodies formed by Low to moderate temperature fluids (70 - 280°C). Meanwhile, they possess complex textures and structures and are predominantly controlled by synsedimentary faults and secondary fractures (Wilkinson, 2003, 2014). In the Xicheng-Fengtai orefield, studies have indicated that carbonate-hosted Zn-Pb deposits, such as the Guojiagou deposit, exhibit a genetic relationship with SEDEX-type deposits (e.g., Changba-Lijiagou, Bijiaoshan) hosted by clastic rocks (Li et al., 1994; Ma et al., 2004; Ma et al., 2007; Zhang et al., 2019b). Syngenetic regional faults and secondary fractures predominantly serve as conduits for fluid pathways rather than storage. The Guojiagou orebodies, situated within Middle Devonian silicified carbonate rocks, have a lenticular morphology and are primarily controlled by lithological and stratigraphic factors. Analysis of C-O isotopes and trace elements suggests that gangue minerals such as ankerite formed in a hydrothermal environment, consistent with the timing of host rock formation (Zhang et al., 2019b). Our detailed field investigations

and micro-petrographic observations at the Guojiagou revealed extensive epigenetic processes including dissolution, brecciation, cementation, infilling, and replacement—rather than syngenetic ore textures—that significantly contribute to Zn-Pb enrichment. Previous S-Pb-C-O isotopic analyses indicate that sulfur originated from the thermal reduction of marine sulfate while metals were sourced from underlying basement rocks; ore fluids derived from the dissolution of host carbonate rocks further substantiate this model (Lin, 1991; Zhang, 2019; Zhang et al., 2019a; Wang, 2020). Extensive research indicates that organic matter not only facilitates thermochemical sulfate reduction but also serves as a chemical barrier to metal-bearing oxidized fluids, enabling the rapid deposition of sulfides nearby (Liao et al., 2020; Guan et al., 2024). Our trace elements analyses provide robust evidence against classifying the Guojiagou Zn-Pb deposit as a magmatic hydrothermal system due to enrichment in low-temperature sensitive trace elements such as Ga and Ge alongside depletion in magmatically associated elements like In and Sn (Seifert & Sandmann, 2006; Pavlova et al., 2015; Chen et al., 2024a; Tan et al., 2024).

In summary, the Guojiagou orebodies display both syngenetic and epigenetic characteristics, and the metallogenetic fluids originate from basin brine rather than magmatism. Based on these observations, we conclude that the Guojiagou deposit belongs to the Irish-type deposits.

We estimate the potential reserves of critical metals Ga and Ge based on our EMPA and LA-ICP-MS trace elements analyses in the Guojiagou Zn-Pb deposit. The total Zn-Pb metal content exceeds 3 Mt, with grades of 3.34% Zn and 1.26% Pb, yielding approximately 2,178,260 t of Zn resources (Zhang et al., 2019a). Combining these figures with mean contents of Ga (68.87 ppm) and Ge (62.76 ppm) in sphalerite from LA-ICP-MS data allows us to calculate the potential reserves of Ga and Ge within the deposit.

Taking Ga as an example, the most precise approach to estimating its reserves in sphalerite within a deposit would involve determining the proportional abundance of various sphalerite types and calculating their respective Ga contributions based on their average contents. This is calculated using the following formula:

$$Ga_{\text{weighted average}} = \sum_{i=0}^n Ga_{i \text{ ave}} * a_i\% \quad (Ga_{\text{weighted average}} - \text{the weighted average amount of Ga across all orebodies, } Ga_{i \text{ ave}} - \text{average content of sphalerite of type } i, a_i\% - \text{percentage of sphalerite of type } i \text{ in all orebodies}), \text{ and } Ga_{\text{reserve}} = \frac{Zn_{\text{resources}} * Ga_{\text{weighted average}}}{Zn_{\text{average}}} \quad (\text{Wu et al., 2023}).$$

However, this method is hindered by the inherent difficulty of accurately quantifying the proportion of each sphalerite type. As an alternative, a more practical approach involves extensive sampling and analysis across multiple stratigraphic levels, averaging the resulting Ga concentrations to approximate the mean Ga content in

sphalerite for the entire deposit. By integrating this average with the mean Zn content in sphalerite and the total Zn metal content, it becomes possible to estimate the Ga reserves. The calculation formula is as follows:

$$Zn_{\text{reserve}} = \frac{(Pb+Zn)_{\text{reserve}} * Zn_{\text{grade}}}{Zn_{\text{grade}} + Pb_{\text{grade}}} = 2178260 \text{ t, and } Ga_{\text{reserve}} = \frac{Zn_{\text{reserve}} * Ga_{\text{average}}}{Zn_{\text{average}}} = 240.07 \text{ t.}$$

Hence, our calculations suggest that the estimated reserves of Ga and Ge within the Guojiagou Zn-Pb deposit may reach up to 240 t for Ga and 219 t for Ge, respectively.

7. Conclusions

(1) Three distinct types of sphalerite at the Guojiagou have been identified: 1) the earliest dark-brown Sp1 intergrown with ankerite; 2) the subsequent brownish-red Sp2 paragenetic with massive pyrite; and 3) the late light-colored Sp3 associated with calcite veins.

(2) Sp3 exhibits significantly high Ga (mean=139.2 ppm) and Ge (mean=160.9 ppm) against Sp1 and Sp2, where Ga enters sphalerite crystal lattice via $Ga^{3+} + Cu^{+} \leftrightarrow 2Zn^{2+}$ and Ge incorporates into the structure as $Ge^{4+} + 2Ag^{+} \leftrightarrow 3Zn^{2+}$ or as $Ge^{2+} \leftrightarrow Cd^{2+}$.

(3) GGIMFis geothermometer indicates that crystallization temperatures decrease from Sp1 and Sp2 (180-230 °C) to Sp3 (120-150 °C), while the presence of ankerite and low Mn concentrations indicate a weakly acidic and relatively oxidizing environment.

(4) The Guojiagou Zn-Pb deposit is estimated to host approximately 240 t of Ga and 219 t of Ge, highlighting the significant potential for both Ga and Ge resources within the Guojiagou Zn-Pb deposit as well as the analogs in the western QOB, China.

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633 LA-ICP-MS analyses.

634 **Table Captions**

635 **Table 1.** Summary statistics of EPMA data for sphalerite (wt. %).

636

637 **Table 2.** Summary statistics of LA-ICP-MS data and temperature estimated results for
638 sphalerite (ppm).

639 **Figure Captions**

640 **Fig. 1.** Geological setting of the Xicheng-Fengtai orefield. (a) Regional geological
641 setting of China: I-Central Asian Orogenic Belt, II-Tarim Craton, III-North China
642 Craton, IV-Central Orogenic Belt, V-Tibetan Plateau, VI-Yangtze Block, VII-
643 Cathaysia Block (b) geological map of Xicheng-Fengtai district and the distribution of
644 Zn-Pb deposits (modified from Xi'an Geological Survey Centre, China Geological
645 Survey). (c) The stratigraphic column of the Devonian in the Xicheng field (modified
646 from [Li et al., 1994](#); [Wang et al., 1996](#); [Zhang, 2019](#)).

647

648 **Fig. 2.** Geological map of the Guojiagou deposit (modified from [Zhang et al., 2019b](#)).

649

650 **Fig. 3.** Cross-sectional map of north ore belt (a) and south ore belt (b) (modified from
651 [Zhang et al., 2019b](#)).

652

653 **Fig. 4.** Photographs showing representative examples of alteration of Guojiagou
654 deposit. (a) The yellow-brown ankerite is interspersed with calcite veins. (b) The
655 crystal cavity structure demonstrates the relationship between ankerite, calcite, and
656 sphalerite. (c) Sphalerite in a spotted form within siliceous limestone. (d) The organic
657 matter is mainly associated with the calcite veins.

658

Fig. 5. Photographs showing representative examples of mineralization of the Guojiagou deposit. (a) Ankerite (Ank0) and calcite (Cc0) occur during the diagenetic stage of the host rock. (b) Framboidal pyrite, sphalerite, and galena (Gn). (c) Black sphalerite (Sp1) replacing ankerite, with dark bands at the contact edges; sphalerite forms a crystal cavity structure with ankerite (Ank1) and calcite (Cc1). (d) Sp1 develops chalcopyrite solid solutions with weak translucency. (e) Ankerite (Ank2) and calcite replace Sp1 + Gn. (f) Pyrite (Py2) accompanies sphalerite (Sp2), replacing Sp1 + Ank1. (g) Py2 develops alongside Sp2 and gradually grows. (h) Sp2 shows enhanced translucency, with less development of chalcopyrite solid solution. (i) Gn forms alongside dolomite. (j) Sp3 fills fractures and enters limestone, interspersed by later calcite veins (Cc4). (k) Sp3 replaces limestone. (l) Sp3 shows good translucency and coexists with calcite veins (Cc3) and ankerite (Ank3).

Fig. 6. Metallogenic stages and mineral paragenetic sequence of the Guojiagou deposit.

Fig. 7. Box and whisker plot for trace elements concentrations of the three-generation sphalerite from the Guojiagou deposit (a-l).

Fig. 8. Representative time-resolved LA-ICP-MS depth profiles for three generations of sphalerite in the Guojiagou deposit (a-c).

Fig. 9. Binary diagrams showing correlations of sphalerite trace elements analyzed by EPMA (a) and LA- ICP- MS (b-i).

Fig. 10. Principal component analysis of the LA–ICP–MS log-transformed dataset of trace elements contents in sphalerite from the Guojiagou deposit. (a) Eigenvalues of the principal components, (b) PCA loading plot showing elements with similar behavior and score plot with histograms of PC1 and PC2 for trace elements compositions of the three sphalerite generations, (c) Loadings of the principal components.

Fig. 11. Simulating the ore-forming environment and enriching Ga and Ge. (a) Boxplots of the calculated temperature of the Guojiagou sphalerite. (b) Ga vs. GGIMF's temperature, (c) Ge vs. GGIMF's temperature, (d) Temperature vs. $\log fS_2$ diagram (Einaudi et al (2003)).

References:

- Bauer, M.E., Burisch, M., Ostendorf, J., Krause, J., Frenzel, M., Seifert, T., Gutzmer, J., 2019. Trace element geochemistry of sphalerite in contrasting hydrothermal fluid systems of the Freiberg district, Germany: insights from LA-ICP-MS analysis, near-infrared light microthermometry of sphalerite-hosted fluid inclusions, and sulfur isotope geochemistry. *Miner. Deposita*. 54 (2), 237-262.
- Belissant, R., Boiron, M., Luais, B., Cathelineau, M., 2014. LA-ICP-MS analyses of minor and trace elements and bulk Ge isotopes in zoned Ge-rich sphalerites from the Noailhac-Saint-Salvy deposit (France): Insights into incorporation mechanisms and ore deposition processes. *Geochim. Cosmochim. Acta*. 126, 518-540.
- Bonnet, J., Mosser-Ruck, R., Caumon, M., Rouer, O., Andre-Mayer, A., Cauzid, J., Peiffert, C., 2016. Trace element distribution (Cu, Ga, Ge, Cd, and Fe) in sphalerite from the Tennessee MVT deposits, USA, by combined EMPA, LA-ICP-MS, Raman spectroscopy, and crystallography. *Can Mineral*. 54, 1261-1284.
- Chen, M.X., Zheng, Y., Chen, X., Yu, P.P., Zhang, G.Z., Wu, Y.H., Huang, Y., Wang, X.H., Shu, L.H., Lin, Z.W., 2024a. High-Cd sphalerite in the Pangxidong Pb-Zn-Ag deposit (Yunkai Domain, South China): Insight for physicochemical condition of orogenic-type deposit. *Ore Geol. Rev.* 167, 105974.
- Chen, X., Wang, Z.K., Yu, P.P., Wu, Y.H., Huang, Y., Tan, Z.J., Zheng, Y., 2024b. Physicochemical condition of the Dafang skarn Au-Ag-Zn-Pb polymetallic deposit in the Nanling Metallogenic Belt (South China), a perspective from sphalerite mineralogy and geochemistry. *Ore Geol. Rev.* 165, 105923.

- 719 Chu, G.B., Chen, H.Y., Zhang, S.T., Zhang, Y., Cheng, J.M., 2023. Geochemistry and
720 geochronology of multi-generation garnet: New insights on the genesis and fluid
721 evolution of prograde skarn formation. *Geosci. Front.* 14(1), 101495.
- 722 Cook, N.J., Ciobanu, C.L., Pring, A., Skinner, W., Shimizu, M., Danyushevsky, L.,
723 Saini-Eidukat, B., Melcher, F., 2009. Trace and minor elements in sphalerite: A
724 LA-ICPMS study. *Geochim. Cosmochim. Acta.* 73, 4761-4791.
- 725 Cooke, D.R., Bull, S.W., Large, R.R., McGoldrick, P.J., 2000. The importance of
726 oxidized brines for the formation of Australian proterozoic stratiform sediment-
727 hosted Pb-Zn (Sedex) deposits. *Econ. Geol.* 95(1), 1-18.
- 728 Cugerone, A., Cenki-Tok, B., Chauvet, A., Le Goff, E., Bailly, L., Alard, O., Allard, M.,
729 2018. Relationships between the occurrence of accessory Ge-minerals and
730 sphalerite in Variscan Pb-Zn deposits of the Bossost anticlinorium, French
731 Pyrenean Axial Zone: Chemistry, microstructures and ore-deposit setting. *Ore*
732 *Geol. Rev.* 95, 1-19.
- 733 Dong, Y.P., Liu, X.M., Neubauer, F., Zhang, G.W., Tao, N., Zhang, Y.G., Zhang, X.N.,
734 Li, W., 2013. Timing of Paleozoic amalgamation between the North China and
735 South China Blocks: evidence from detrital zircon U-Pb ages. *Tectonophysics.* 586,
736 173-191.
- 737 Dong, Y.P., Santosh, M., 2016. Tectonic architecture and multiple orogeny of the
738 Qinling Orogenic Belt, Central China. *Gondwana Res.* 29(1), 1-40.
- 739 Einaudi, M.T., Hedenquist J.W., Inan, E.E., 2003. Sulfidation state of fluids in active
740 and extinct hydrothermal systems: transition from porphyry to epithermal
741 environments. In: Simmons SF, Graham (eds) *Volcanic, Geothermal, and*
742 *OreForming Fluids: Rulers and Witnesses of Processes within the Earth.* Society
743 of Economic Geologists and Geochemical Society, Special Publication. 10, 285–
744 313.
- 745 Fang, W.X., 2023. Hydrothermal sedimentary lithofacies and their tectono-
746 palaeogeography. *J. Palaeogeogr.* 25(3), 1-29 in Chinese with English abstract.
- 747 Frenzel, M., Hirsch, T., Gutzmer, J., 2016. Gallium, germanium, indium, and other trace
748 and minor elements in sphalerite as a function of deposit type; a meta-analysis.
749 *Ore Geol. Rev.* 76, 52-78.
- 750 Frenzel, M., Voudouris, P., Cook, N.J., Ciobanu, C.L., Gilbert, S., Wade, B.P., 2021.
751 Evolution of a hydrothermal ore-forming system recorded by sulfide mineral
752 chemistry: a case study from the Plaka Pb-Zn-Ag Deposit, Lavrion, Greece.
753 *Mineral. Deposita.* 57, 417-438.

- 754 Gao, Y.B., Li, K., Qian, B., Li, W.Y., Zheng, M.C., Zhang, C.G., Trace elements, S, Pb,
755 He, Ar and C isotopes of sphalerite in the Mayuan Pb-Zn deposit, at the northern
756 margin of the Yangtze plate, China, *Acta Petrologica Sinica*. 32(01), 251-263 in
757 Chinese with English abstract.
- 758 Guan, G.Y., Li, S.W., Li, R.X., W, W., 2024. Impact of temperature on sphalerite
759 chemistry: Simulation experiments of sphalerite in MVT Zn-Pb sulfide deposits
760 in the Beiba Dome, northern Sichuan Basin, China. *Ore Geol. Rev.* 174, 106325.
- 761 Guo, A.Q., Ding Z.J., Wen C.X., 2011. Zircon U-Pb dating of the granite in Xicheng
762 orefield, Gansu Province and its geological significance. *Acta Mineralogica Sinica*.
763 (A1), 581-582 in Chinese.
- 764 Guo, D.B., Bai, R.L., Dai, S., Chen, X.F., Niu, Y.J., Shang, L.L., Ali, M.M., Guo, X.G.,
765 Zhang, C.X., Wang, B.L., 2023. Trace elements of sulfides in the Dengjiashan Pb-
766 Zn deposit from West Qinling, China: Implications for mineralization conditions
767 and genesis. *Geol. J.* 58, 2913-2938.
- 768 Han, Z.X., 1994. The typomorphic characteristic of the sphalerite in the Qinling
769 Devonian system lead-zinc metallogenic belt. *Journal of Xi'an Institute of*
770 *Geology*. 16(1), 12-17 in Chinese.
- 771 Hu, Q.Q., Wang, Y.J., Mao, J.W., Wei, R., Liu, S.Y., Ye, D., Yuan, Q.H., Dou, P., 2015.
772 Timing of the formation of the Changba-Lijiagou Pb-Zn ore deposit, Gansu
773 Province, China: Evidence from Rb-Sr isotopic dating of sulfides. *J. Asian Earth*
774 *Sci.* 103, 350-359.
- 775 Hu, Q.Q., Wang, Y.T., Mao, J.W., Liu, X.L., Chen, S.C., Wei, R., Zhang, J., Wang, R.T.,
776 Wang, C.G., Dai, J.Z., Wen, S.W., Chen, M.S., 2020. Genesis of the Bafangshan-
777 Erlihe Zn-Pb-Cu deposit in the Fengxian-Taibai ore cluster, west Qinling, China:
778 Evidence from ore geology and ore-forming fluids. *Ore Geol. Rev.* 126, 103734.
- 779 Hu, Q.Q., Wang, Y.T., Chen, S.C., Wei, R., Liu, X.L., Liu, J.C., Wang, R.T., Gao, W.H.,
780 Wang, C.G., Tang, M.J., Wu, W.T., 2024. Genesis of the Dongtangzi Zn-Pb
781 Deposit of the Fengxian-Taibai Ore Cluster in West Qinling, China: Constraints
782 from Rb-Sr and Sm-Nd Geochronology, and In Situ S-Pb Isotopes. *Minerals*. 14,
783 297.
- 784 Keller, P.C., Anderson, C.G., 2018. The production of critical materials as by products.
785 *Asp. min. miner. sci.* 2(2), 208-221.
- 786 Kelley, K.D., Leach, D.L., Johnson, C.A., Clark, J.L., Fayek, M., Slack, J.F., Anderson,
787 V.M., Ayuso, R.A., Ridley, W.I., 2004. Textural, compositional, and sulfur isotope
788 variations of sulfide minerals in the Red Dog Zn-Pb-Ag deposits, Brooks Range,
789 Alaska: implications for ore formation. *Econ. Geol.* 99(7), 1509-1532.

- 790 Lai, S.C., Qin, J.F., Chen, L., Grapes, R., 2010. Geochemistry of Ophiolites from the
791 Mian-Lue Suture Zone: Implications for the Tectonic Evolution of the Qinling
792 Orogen, Central China. *Int Geol Rev.* 50, 650-664.
- 793 Leach, D.L., Song, Y., 2019. Sediment-hosted zinc-lead and copper deposits in China,
794 in Chang, Z.S., Goldfarb, R.J. (eds.), *Mineral. Deposits of China: Society of*
795 *Economic Geologists, Special Publication 22*, pp. 325–409.
- 796 Li, C.Y., Wang, F.Y., Hao, X.L., Ding, X., Zhang, H., Ling, M.X., Zhou, J.B., Li, Y.L.,
797 Fan, W.M., Sun, W.D., 2012. Formation of the world's largest molybdenum
798 metallogenic belt: a plate tectonic perspective on the Qinling molybdenum
799 deposits. *Int Geol Rev.* 54, 1093-1112.
- 800 Li, Z.C., P, X.Z., Li, R.B., Pei, L., H, B., Liu, C.J., Chen, G.C., Chen, Y.X., 2013. LA-
801 ICP-MS zircon U-Pb dating, geochemistry of the Mishuling intrusion in western
802 Qin ling and their tectonic significance. *Acta Petrologica Sinica.* 29(8), 2617-2634
803 in Chinese with English abstract.
- 804 Li J.S., Cao, X.D., Yang J.L., Yin, H.F., Yang, F.Q., Zhang, R.L., Hu, Y.X., Lai, X.L.,
805 Yang, Z.Y., Yu, S.Y., Zhu, H.Y., Lai, C.G., Xiao, J.D., Chen, Y.N., 1994. The
806 sedimentation and evolutionary history of Phanerozoic Ocean Basin in the Qinling.
807 Beijing: Geological Publishing House, 1-206 in Chinese.
- 808 Li, S.Z., Kushy, T.M., Wang, L., Zhang, G.W., Lai, S.C., Liu, X.C., Dong, S.W., Zhao,
809 G.C., 2007. Collision leading to multiple-stage large-scale extrusion in the qinling
810 orogen: Insights from the Mianlue suture. *Gondwana Res.* 12(1-2), 121-143.
- 811 Liao, J., Wang, T.L., Lu, H., Greenwood, P.F., Peng, P.A., Hsu, C.S., 2020. Effects of
812 Mississippi valley-type minerogenetic metal sulfates on thermochemical sulfate
813 reduction, studied by hydrous pyrolysis. *Org. Geochem.* 150, 104128-104140.
- 814 Lin, B., 1991. Metallogenetic information on the lead isotopic geochemistry of the
815 Xicheng lead-zinc ore field, Gansu Province. *Geo sciences.* 5, 300-306 in Chinese
816 with English abstract.
- 817 Liu, T.G., Ye, L., Zhou, J.X., Wang, X.L., 2010. Cd Primarily Isomorphously Replaces
818 Fe but not Zn in Sphalerite. *Acta Mineralogica Sinica.* 30(2), 179-184 in Chinese
819 with English abstract.
- 820 Liu, Y.J., Cao, L.M., Li, Z.L., Wang, H.N., Chu, T.Q., Zhang, J.R., 1984. Geochemistry
821 of element. Beijing: Science Press, 1-548 in Chinese.
- 822 Liu, Y.S., Hu, Z.C., Gao, S., Günther, D., Xu, J., Gao, C.G., Chen, H.H., 2008. In situ
823 analysis of major and trace elements of anhydrous minerals by LA-ICP-MS
824 without applying an internal standard. *Chem. Geol.* 257, 34-43.

- 825 Lockington, J.A., Cook, N.J., Ciobanu, C.L., 2014. Trace and minor elements in
826 sphalerite from metamorphosed sulphide deposits. *Miner. Petrol.* 108, 873-890.
- 827 Ma, G.L., Beaudoin, G., Qi, S.J., Li, Y., 2004. Geology and geochemistry of the
828 Changba SEDEX Pb-Zn deposit, Qinling orogenic belt, China. *Mineral. Deposita.*
829 39, 380-395.
- 830 Ma, G.L., Beaudoin, G., Zhong, S.J., Li, Y., Zeng, Z.R., 2007. Geology and
831 geochemistry of the Dengjiashan Zn-Pb SEDEX deposit, Qinling Belt, China:
832 Canadian. *J. Earth Sci.* 44, 479-492.
- 833 Mao, J.W., Qiu, Y.M., Goldfarb, R.J., Zhang, Z.Z., Garwin, S., Shou, R.F., 2002.
834 Geology, distribution, and classification of gold deposits in the western Qinling
835 Belt, central China. *Mineral. Deposita.* 37, 352-377.
- 836 Maslennikov, V.V., Maslennikova, S.P., Large, R.R., Danyushevsky, L.V., 2009. Study
837 of trace element zonation in vent chimneys from the Silurian Yaman-Kasy
838 volcanichosted massive sulfide deposit (Southern Urals, Russia) using laser
839 ablationinductively coupled plasma mass spectrometry (LA-ICPMS). *Econ. Geol.*
840 104 (8), 1111-1141.
- 841 Meng, Y.M., Zhang, X., Huang, X.W., Hu, R.Z., Bi, X.W., Meng, S.N., Zhou, L.L.,
842 Zheng, Y., 2024. A review of the Zn-Pb deposits in Sichuan-Yunnan-Guizhou
843 metallogenic region with emphasis on the enrichment mechanism of Ge, Ga, and
844 In. *Ore Geol. Rev.* 164, 105853.
- 845 Möller, P., 1987. Correlation of homogenization temperatures of accessory minerals
846 from sphalerite-bearing deposits and Ga/Ge model temperatures. *Chem. Geol.*
847 61(1-4), 153-159.
- 848 Murakami, H., Ishihara, S., 2013. Trace elements of indium-bearing sphalerite from
849 tinpolymetallic deposits in Bolivia, China and Japan: a femto-second LA-ICPMS
850 study. *Ore Geol. Rev.* 53, 223-243.
- 851 Ni, P., Wang, T.G., Wang, G.G., Li, W.S., Pan, J.Y., 2019. Metamorphic fluid
852 superimposition of the Changba-Lijiagou Pb-Zn deposit, West Qinling Orogen,
853 central China. *Metamorphic Geology: Microscale to Mountain Belts.* Geological
854 Society, London, Special Publications. 478, 265-285.
- 855 Paton, C., Hellstrom, J., Paul, B., Woodhead, J., Hergt, J., 2011. Iolite: Freeware for the
856 visualisation and processing of mass spectrometric data. *J. Anal. At. Spectrom.* 26,
857 2508.
- 858 Pavlova, G.G., Palessky, S.V., Borisenko, A.S., Vladimirov, A.G., Seifert, T., Phan,
859 L.A., 2015. Indium in cassiterite and ores of tin deposits. *Ore Geol. Rev.* 66, 99-

- 860 113.
- 861 Pu, F.C., 2018. Geological Feature and Genetic Types of Guojiagou Lead-Zinc Mine in
862 Huixian County of Gansu Province. *Gansu Metallurgy*. 40(4), 38-40 in Chinese
863 with English abstract.
- 864 Qiu, K.F., Deng, J., Yu, H.C., Rasbury, T., Tang, Y.W., Zhu, R., Zhang, P.C., Goldfarb,
865 R., 2024. The Zaozigou orogenic gold-antimony deposit, West Qinling Orogen,
866 China: Structural controls on multiple mineralization events: *Geological Society
867 of America Bulletin*. 136 (9-10), 4218-4232.
- 868 Qiu, K.F., Deng, J., He, D.Y., Rosenbaum, G., Zheng, X., Williams-Jones, A.E., Yu,
869 H.C., Balen, D., 2023. Evidence of vertical slab tearing in the Late Triassic Qinling
870 Orogen (central China) from multiproxy geochemical and isotopic imaging.
871 *Journal of Geophysical Research: Solid Earth*. 128, e2022JB025514.
- 872 Qiu, K.F., Yu, H.C., Hetherington, C., Huang, Y.Q., Yang, T., Deng, J., 2021.
873 Tourmaline composition and boron isotope signature as a tracer of magmatic-
874 hydrothermal processes. *American Mineralogist*. 106, 1033-1044.
- 875 Qiu, K.F., Yu, H.C., Deng, J., McIntire, D., Gou, Z.Y., Geng, J.Z., Chang, Z.S., Zhu, R.,
876 Li, K.N., Goldfarb, R.J., 2020. The giant Zaozigou orogenic Au-Sb deposit in West
877 Qinling, China: Magmatic or metamorphic origin? *Mineralium Deposita*. 55(2),
878 345-362.
- 879 Schulz, K.J., 2017. Critical mineral resources of the United States: economic and
880 environmental geology and prospects for future supply, Geological Survey.
- 881 Seifert, T., Sandmann, D., 2006. Mineralogy and geochemistry of indium-bearing
882 polymetallic vein-type deposits: Implications for host minerals from the Freiberg
883 district, eastern Erzgebirge Germany. *Ore Geol. Rev.* 28(1), 1-31.
- 884 Shu, L.H., Chen, X., Yu, P.P., Huang, Y., Wang, X.H., Li, R.J., Tan, Z.J., Chen, M.X.,
885 Zheng, Y., 2024. Oxidative fluids facilitate Cd enrichment in a magmatic-
886 hydrothermal system: An example from the Kangjiawan Pb-Zn polymetallic
887 deposit, South China. *Ore Geol. Rev.* 165, 105926,
- 888 Tan, Z.J., Zheng, Y., Yu, P.P., Li, R.J., Huang, Y., Ding, W., Wu, Y.H., Chen, M.X., Sun,
889 L.H., Wang, Z.K., Chen, X., 2024. Indium concentrations as a potential indicator
890 of orebody-intrusion distance in a skarn Pb-Zn system, elucidated by the
891 Fozichong Orefield (South China). *Ore Geol. Rev.* 165, 105912.
- 892 Wang J.L., He B.C., Li, J.Z., He, D.R., 1996. Qinling Type Lead and Zinc Mineral
893 Deposits in China. Beijing: Geological Publishing House. 1-264 in Chinese with
894 English abstract.

- 895 Wang, J., Yu, H.C., Petrella, L., Duo, D.W., Wang, L., Yang, F., Zhou, H.Y., Li, C., Qiu,
896 K.F., 2024a. Fluid evolution and genesis of the Yidinan granitoid-hosted orogenic
897 gold deposit (China). *Geological Society of America Bulletin*.
- 898 Wang, L., Zhang, Y., Han, R.S., Li, X.D., 2023. LA-ICP-MS analyses of trace elements
899 in zoned sphalerite: A study from the Maoping carbonate-hosted Pb-Zn(-Ge)
900 deposit, southwest China. *Ore Geol. Rev.* 157, 105468.
- 901 Wang, Q., 2020. Analysis of metallogenic materials of lead-zinc deposits in Xicheng
902 ore field, Gansu Province. *World Nonferrous Metals*. (09), 109-110 in Chinese
903 with English abstract.
- 904 Wang, R.T., Li, F.L., Chen, E.H., Dai, J.Z., Wang, C.A., Xu, X.F., 2011. Geochemical
905 characteristics and prospecting prediction of the Bafangshan-Erlihe large lead-zinc
906 ore deposit, Feng County, Shaanxi Province, China. *Acta Petrol. Sin.* 27(3), 779-
907 793 in Chinese with English abstract.
- 908 Wang, R.T., Zhang, G.L., Li, Q.F., Zhang, B., Cheng, H., Ji, Y.F., 2021. Metallogenic
909 Regularity and Prospecting Prediction of Fengtai Pb-Zn-Au Ore Concentration
910 Area in Qinling, China. *Journal of Earth Sciences and Environment*. 43(3), 528-
911 548.
- 912 Wang, X.H., Zheng, Y., Yu, P.P., Chen, X., Wu, Y.H., Huang, Y., Long, L.J., Shu, L.H.,
913 Chen, M.X., Guo, L.X., 2024b. Temperature as a major control on Cd enrichment
914 in a skarn system: A case study of the Yiliu Pb-Zn-As deposit, South China. *Ore*
915 *Geol. Rev.* 165, 105920.
- 916 Wei, R., Wang, Y.T., Hu, Q.Q., Huang, S.K., Yuan, Q.H., Ba, Q.L., Hu, W.R., Zhang,
917 X.J., Cai, T., 2017. Zircon U-Pb ages and Hf isotope compositions of Changba
918 and Huangzhuguan plutons in West Qinling, and their geological significance.
919 *Mineral. Deposita*. 36(6), 1367-1386 in Chinese with English abstract.
- 920 Wen, H.J., Zhu, C.W., Du, S.J., Fan, Y., Luo, C.G., 2020. Gallium (Ga), germanium
921 (Ge), thallium (Tl) and cadmium (Cd) resources in China. *Chin Sci Bull.* 65, 3688-
922 3699 in Chinese with English abstract.
- 923 Wilkinson, J.J., 2003. On diagenesis, dolomitisation and mineralisation in the Irish Zn-
924 Pb orefield. *Miner Deposita* 38, 968 - 983. [https://doi.org/10.1007/s00126-003-](https://doi.org/10.1007/s00126-003-0387-7)
925 0387-7.
- 926 Wilkinson, J.J., 2014. Sediment-Hosted Zinc-Lead mineralization: Processes and
927 perspectives. In: Holland, H.D, Turekian, K.K. (Eds.), *Treatise on Geochemistry*
928 (Second Edition). Oxford, Elsevier, 219-249.
- 929 Wilson, S.A., Ridley, W.I., Koenig, A.E., 2002. Development of sulfide calibration

- standards for the laser ablation inductively-coupled plasma mass spectrometry technique. *J. Anal. At. Spectrom.* 17, 406-409.
- Wu, Y.H., Yu, P.P., Chen, X., Hu, Z.B., Long, L.J., He, B., Zheng, Y., 2023. Earlier stage, higher temperature, and deeper space facilitate indium precipitation in a skarn system, as exemplified by the Baoshan Pb-Zn polymetallic deposit, South China: *Ore Geol. Rev.* 163, 105745.
- Wu, Y.H., Zheng, Y., Yu, P.P., Chen, X., Guo, L.X., Long, L.J., Hu, Z.B., Huang, Y., Chen, M.X., Lu, J.J., Ouyang, S.Y., Chen, S.Z., 2024. The world-class carbonate-hosted Fankou Zn-Pb deposit in China. Part III. Critical metal Ga is enriched in late stage and low temperature sphalerite. *Ore Geol. Rev.* 166, 105928.
- Ye, L., Cook, N.J., Ciobanu, C.L., Liu, Y.P., Zhang, Q., Liu, T.G., Gao, W., Yang, Y.L., Danyushevskiy, L., 2011. Trace and minor elements in sphalerite from base metal deposits in South China: A LA-ICPMS study. *Ore Geol. Rev.* 39, 188-217.
- Ye, L., Gao, W., Yang, Y.L., Liu, T.G., Peng, S.S., 2012. Trace elements in sphalerite in Laochang Pb-Zn polymetallic deposit, Lancang, Yunnan Province. *Acta Petrologica Sinica*, 28(5), 1362-1372 in Chinese with English abstract.
- Yu, C.W., Shen, Y.L., Cen, K., Zhou, D.Y., Su, C.M., Huang, Y.J., 1987. Research on Regional Geochemistry in Yiliu Region, Guangdong Province. Geological Publishing House Beijing, 202 in Chinese.
- Yu, H.C., Qiu, K.F., Deng, J., Zhu, R., Mathieu, L., Sai, S.X., Sha, W. J., 2022a. Exhuming and preserving epizonal orogenic Au-Sb deposits in rapidly uplifting orogenic settings. *Tectonics*, 41, e2021TC007165.
- Yu, H.C., Qiu, K.F., Pirajno, F., Zhang, P.C., Dong, W.Q., 2022b. Revisiting Phanerozoic evolution of the Qinling Orogen (East Tethys) with perspectives of detrital zircon. *Gondwana Research*, 103, 426-444.
- Yu, H.C., Qiu, K.F., Hetherington, C.J., Chew, D., Huang, Y.Q., He, D.Y., Geng, J.Z., Xian, H.Y., 2021. Apatite as an alternative petrochronometer to trace the evolution of magmatic systems containing metamict zircon. *Contributions to Mineralogy and Petrology*, 176, 68.
- Yu, P.P., Hu, Z.B., Zheng, Y., Guo, L.X., Ouyang, S.Y., Yuan, G.Q., Chen, S.Z., Wang, C.M., Chen, X., 2024. A potential giant gallium deposit hosted in the tailing dam of the Fankou Zn-Pb deposit in northern Guangdong Province, South China: *Geol. China*. 7, 157-160.
- Yu, Z.H., Zhu, X.Y., Tong, S.Y., Song, J.Y., Wang, D.B., Wei, Z.G., 2008. Study on characteristics of S, Pb isotopes of the Pb-Zn deposits, Au deposits in Xicheng

- 965 area and their metallogenic relationship. *Mineral Resources and Geology*. 22(3),
966 196-205 in Chinese with English abstract.
- 967 Yuan, X.Y., Niu, M.L., Cai, Q.R., Zhu, G., Wu, Q., Li, X.C., Sun, Y., Li, C., and Qian,
968 T., 2022. The nature of Paleoproterozoic basement in the northern Yangtze and its
969 geological implication. *Precambrian Res.* 378, 106761.
- 970 Zhang, B.Q., Ding, Z.J., Cai, H.G., Wang, Z.X., 2015. Mineralization characteristic of
971 Guojiagou lead-zinc deposit. *Modern Mining*. (8), 87-90 in Chinese with English
972 abstract.
- 973 Zhang, F., Liu, S.W., Li, Q.G., Sun, Y.L., Wang, Z.Q., Yan, Q.R., Yan, Z., 2011. Re-Os
974 and U-Pb geochronology of the Erlihe Pb-Zn deposit, Qinling orogenic belt,
975 central China, and constraints on its deposit genesis. *ACTA GEOL SIN-ENGL.* 85,
976 673-682.
- 977 Zhang, G.W., Dong, Y.P., Lai, S.C., Guo, A.L., Meng, Q.R., Liu, S.F., Cheng, S.Y., Yao,
978 A.P., Zhang, Z.Q., Pei, X.Z., Li, S.Z., 2004. Mianlue tectonic zone and Mianlue
979 suture zone on southern margin of Qinling-Dabie orogenic belt. *SCI CHINA SER*
980 *D.* 47, 300-316.
- 981 Zhang, H., He, J., Li, Y.C., Wu, G.H., Cheng, H., Chen, F.K., 2023. Neoproterozoic
982 crustal growth and reworking in South Qinling, central China: Evidence from
983 zircon U-Pb age and Hf isotopic composition of paragneiss in the Foping dome. *J.*
984 *Asian Earth Sci.* 256, 105784.
- 985 Zhang, S.X., 2019. Study on Prospecting Model and Metallogenic Prediction of Hidden
986 Lead-zinc Deposit in Xicheng Ore Field (PhD Thesis). Ph. D. Dissertation. Wuhan:
987 China University of Geosciences, 1-219 in Chinese with English abstract.
- 988 Zhang, S.X., Ding, Z.J., Cai, H.A., Zhang, B.Q., Wang, Z.X., Li, Z.J., Liu, Z.Y., 2019a.
989 Genesis of the Guojiagou Pb-Zn deposit in Huixian County, Gansu Province: ore-
990 bearing siliceous rock constant, rare earth elements and ore isotopic evidence.
991 *Bulletin of Geological Science and Technology*. 38(4), 93-107 in Chinese with
992 English abstract.
- 993 Zhang, S.X., Hu, Q.Q., Wang, Y.T., Wei, R., Ke, C.H., 2019b. Characteristics of ore
994 geology and ore-controlling factors of giant Guojiagou Pb-Zn deposit in Xicheng
995 ore concentration area, western Qinling. *Mineral. Deposita*. 38(5), 1129-1146 in
996 Chinese with English abstract.
- 997 Zhao, T., Qin, P.Z., Wang, A.J., Wang, G.S., Li, J.W., Liu C., Liu, Y.F., 2017a. An
998 Analysis of Gallium Ore Resources Demand Trend and the Thinking concerning
999 China's Gallium Industry Development. *Acta Geoscientica Sinica*. 38(01), 77-84
1000 in Chinese with English abstract.

- 1001 Zhao, Y., Zhai, M.G., Chen, H., Zhang S.H., 2017b. Paleozoic-early Jurassic tectonic
1002 evolution of North China Craton and its adjacent orogenic belts. *Geol. China* 44(1),
1003 44-60.
- 1004 Zheng, B.S., Mou, C.L., Wang, X.P., Chen, H.D., 2021. U-Pb ages, trace elements and
1005 Hf isotopes of detrital zircons from the late Permian-early Triassic sedimentary
1006 succession in the northern Yangtze Block, South China: Implications for the
1007 reconstruction of the South China Block in Pangea. *J. Asian Earth Sci.* 206,
1008 104609.
- 1009 Zheng, Y., Yu, P.P., Li, Z.K., Xiong, S.F., Zhou, L.L., Zhou, J.X., Wang, C.M., Meng,
1010 Y.M., Zhang, Y., Wang, Y.J., Xu, J., Wu, Y., Guo, L.X., Zhao, T.P., 2023. Critical
1011 Metals Ga, Ge and In in the Global Pb-Zn Deposits: Current Understanding,
1012 Challenges and Perspectives. *J. Earth Sci.* 34(4), 1308-1311.

Highlights

- 1013
- 1014 ➤ Three sphalerite generations at Guojiagou show varied colors and rising Ga-Ge.
- 1015 ➤ Ga and Ge enrich in late-stage, low-temperature, low-sulfur fugacity sphalerite.
- 1016 ➤ Geological and geochemical data classify Guojiagou as an Irish-type deposit.
- 1017 ➤ Over 240 t Ga and 219 t Ge are estimated to be at the Guojiagou deposit.

1018

Appendix Table 1. Summary statistics of EPMA data for sphalerite (wt. %)

Samples No.	Ge	Ga	Cu	Fe	Mn	S	Cd	In	Sn	Zn	Total
Sp1 n = 25											
	0.0	0.0	0.0	3.2	0.0	32.8	0.3	0.0	0.0	62.7	99.3
GB-1-1-SP@1	0	0	0	9	1	6	9	0	0	8	2
	0.0	0.0	0.0	3.3	0.0	32.5	0.3	0.0	0.0	62.8	99.1
GB-1-1-SP@2	0	0	0	0	1	9	8	0	0	7	5

GB-1-1-SP@3	0.0 0	0.0 0	0.0 0	3.1 8	0.0 1	32.6 9	0.3 7	0.0 0	0.0 0	62.3 6	98.6 0
GB-1-1-SP@4	0.0 0	0.0 0	0.0 0	2.9 4	0.0 0	32.5 7	0.3 5	0.0 0	0.0 0	63.3 1	99.1 6
GB-1-1-SP@5	0.0 0	0.0 0	0.0 0	2.9 4	0.0 0	32.5 8	0.3 5	0.0 0	0.0 0	63.1 1	98.9 8
GB-1-1-SP@6	0.0 0	0.0 0	0.0 0	3.2 1	0.0 0	32.7 4	0.3 8	0.0 0	0.0 0	62.8 5	99.1 8
GJG-1-13-1@1	0.0 0	0.0 0	0.0 0	3.4 7	0.0 0	32.5 2	0.2 9	0.0 1	0.0 0	62.7 2	99.0 1
GJG-1-13-1@2	0.0 0	0.0 0	0.0 0	3.1 4	0.0 0	32.6 3	0.2 8	0.0 0	0.0 0	62.4 5	98.5 0
GJG-1-13-1@3	0.0 0	0.0 0	0.0 0	3.3 1	0.0 0	32.5 4	0.2 5	0.0 0	0.0 0	62.7 3	98.8 3
GJG-1-13-1@4	0.0 0	0.0 0	0.0 0	3.2 3	0.0 1	32.5 6	0.2 6	0.0 0	0.0 0	62.9 2	98.9 7
GJG-1-13-1@5	0.0 0	0.0 0	0.0 0	3.2 4	0.0 0	32.2 9	0.2 5	0.0 0	0.0 2	62.8 2	98.6 1
GJG-1-13-1@6	0.0 0	0.0 0	0.0 0	3.0 8	0.0 0	32.5 9	0.2 6	0.0 0	0.0 0	62.7 4	98.6 6
GJ-N7-22-1@1	0.0 0	0.0 0	0.1 8	1.4 3	0.0 1	32.3 3	0.3 8	0.0 0	0.0 0	64.7 1	99.0 5
GJ-N7-22-1@2	0.0 0	0.0 0	0.2 3	1.4 1	0.0 0	32.4 0	0.3 3	0.0 0	0.0 0	64.8 5	99.2 2

GJ-N7-22-1@3	0.0 0	0.0 0	0.6 0	1.5 3	0.0 0	32.6 6	0.3 5	0.0 1	0.0 0	64.3 5	99.5 0
GJ-N7-22-1@4	0.0 0	0.0 0	0.3 2	1.2 6	0.0 0	32.9 9	0.3 4	0.0 0	0.0 1	64.8 2	99.7 3
GJ-N7-22-1@5	0.0 0	0.0 0	0.6 8	1.3 9	0.0 0	32.2 7	0.3 7	0.0 0	0.0 1	64.1 8	98.8 9
GJ-N7-22-1@6	0.0 0	0.0 0	0.5 4	1.1 1	0.0 0	32.5 6	0.4 8	0.0 0	0.0 0	64.6 5	99.3 5
GT-N7-18-Z-1@1	0.0 0	0.0 0	0.0 0	3.3 6	0.0 2	32.8 5	0.3 0	0.0 0	0.0 1	62.9 2	99.4 4
GT-N7-18-Z-1@2	0.0 0	0.0 0	0.2 8	3.1 7	0.0 0	32.4 3	0.2 9	0.0 0	0.0 0	63.4 0	99.5 6
GT-N7-18-Z-1@3	0.0 0	0.0 0	0.0 0	3.6 6	0.0 0	32.6 2	0.3 5	0.0 0	0.0 0	62.5 2	99.1 6
GT-N7-18-Z-1@4	0.0 0	0.0 0	0.1 1	2.6 8	0.0 1	32.1 4	0.3 8	0.0 0	0.0 0	63.1 8	98.4 9
GJ-1-3-2@1	0.0 0	0.0 0	0.0 1	3.9 8	0.0 0	32.7 1	0.3 3	0.0 0	0.0 0	62.2 2	99.2 5
GJ-1-3-2@2	0.0 0	0.0 0	0.0 5	3.4 8	0.0 1	32.7 5	0.3 1	0.0 0	0.0 0	62.9 4	99.5 4
GJ-1-3-2@3	0.0 0	0.0 0	0.0 0	3.0 9	0.0 0	32.4 8	0.2 7	0.0 1	0.0 0	62.7 0	98.5 4

Sp2 n= 31

GJ-1-3-2@4	0.0 0	0.0 0	0.0 0	5.0 5	0.0 0	32.2 7	0.3 0	0.0 0	0.0 0	61.2 2	98.8 4
GJ-1-3-2@5	0.0 0	0.0 0	0.0 0	4.6 7	0.0 0	32.5 2	0.3 9	0.0 0	0.0 0	61.9 5	99.5 3
GJ-1-3-2@6	0.0 0	0.0 0	0.0 0	4.7 2	0.0 0	32.5 1	0.3 9	0.0 0	0.0 0	61.3 9	99.0 1
GT-N7-18-Z- 1@5	0.0 0	0.0 0	0.0 4	2.9 8	0.0 0	32.8 8	0.2 5	0.0 0	0.0 0	63.0 8	99.2 3
GT-N7-18-Z- 1@6	0.0 0	0.0 0	0.0 1	2.9 4	0.0 1	32.0 1	0.2 8	0.0 0	0.0 0	63.2 6	98.5 0
GJG-1-36J-1- SP@1	0.0 0	0.0 0	0.0 0	3.4 0	0.0 1	32.2 8	0.3 5	0.0 0	0.0 1	62.7 3	98.7 7
GJG-1-36J-1- SP@2	0.0 0	0.0 0	0.0 0	3.3 7	0.0 0	32.4 4	0.3 3	0.0 0	0.0 0	62.1 4	98.2 9
GJG-1-36J-1- SP@3	0.0 0	0.0 0	0.0 0	3.3 3	0.0 0	32.5 8	0.3 4	0.0 0	0.0 0	62.3 6	98.6 2
GJG-1-36J-1- SP@4	0.0 0	0.0 0	0.0 0	3.4 1	0.0 0	32.1 9	0.3 4	0.0 0	0.0 1	63.1 9	99.1 3
GJG-1-36J-1- SP@5	0.0 0	0.0 0	0.0 0	3.3 7	0.0 1	32.4 6	0.3 5	0.0 0	0.0 0	62.5 7	98.7 6
GJG-1-36J-1- SP@6	0.0 0	0.0 0	0.0 0	3.3 7	0.0 1	32.5 2	0.3 6	0.0 0	0.0 0	63.1 8	99.4 3
GJG-1-5-1-SP@1	0.0 0	0.0 0	0.0 0	3.4 5	0.0 1	32.5 6	0.3 5	0.0 0	0.0 1	62.9 2	99.3 0

	0.0	0.0	0.0	3.5	0.0	32.6	0.3	0.0	0.0	63.1	99.7
GJG-1-5-1-SP@2	0	0	0	7	0	3	6	0	2	9	7

(continued on next page)

Table 1. (continued)

Samples No.	Ge	Ga	Cu	Fe	Mn	S	Cd	In	Sn	Zn	Total
	0.0	0.0	0.0	3.6	0.0	32.6	0.3	0.0	0.0	62.4	99.2
GJG-1-5-1-SP@3	0	0	7	1	1	2	6	0	6	7	1
	0.0	0.0	0.0	3.6	0.0	32.5	0.3	0.0	0.0	62.9	99.5
GJG-1-5-1-SP@4	0	0	0	2	0	8	6	0	1	3	0
	0.0	0.0	0.0	3.6	0.0	32.7	0.3	0.0	0.0	63.0	99.8
GJG-1-5-1-SP@5	0	0	0	7	0	8	7	0	1	0	2
	0.0	0.0	0.0	3.7	0.0	32.6	0.3	0.0	0.0	62.5	99.2
GJG-1-5-1-SP@6	0	0	1	4	0	1	8	0	1	3	6
	0.0	0.0	0.0	2.5	0.0	32.5	0.2	0.0	0.0	63.7	99.1
GJ-S1-11-1@1	0	0	0	3	0	9	6	0	0	6	4
	0.0	0.0	0.0	2.8	0.0	32.3	0.2	0.0	0.0	63.2	98.7
GJ-S1-11-1@2	0	0	0	4	0	8	6	0	1	1	0
	0.0	0.0	0.0	4.2	0.0	32.4	0.2	0.0	0.0	61.8	98.7
GJ-S1-11-1@3	0	0	0	4	0	3	9	0	0	3	8
	0.0	0.0	0.0	3.2	0.0	32.7	0.2	0.0	0.0	63.0	99.4
GJ-S1-11-1@4	0	0	0	5	0	9	9	0	1	8	1

GJ-S1-11-1@5	0.0 0	0.0 0	0.0 0	4.7 5	0.0 0	32.3 5	0.3 3	0.0 0	0.0 2	61.5 5	98.9 9
GJ-S1-11-1@6	0.0 0	0.0 0	0.0 0	4.6 3	0.0 0	32.2 6	0.3 4	0.0 0	0.0 0	61.6 5	98.8 7
GTG-1-2-1L@1	0.0 0	0.0 0	0.0 4	3.4 8	0.0 0	32.6 9	0.2 6	0.0 0	0.0 0	62.8 3	99.3 0
GTG-1-2-1L@2	0.0 0	0.0 0	0.0 1	4.3 3	0.0 0	32.4 8	0.3 4	0.0 0	0.0 0	61.6 5	98.8 1
GTG-1-2-1L@3	0.0 0	0.0 0	0.0 0	4.7 8	0.0 0	32.4 2	0.3 7	0.0 0	0.0 0	61.4 3	99.0 0
GTG-1-2-1L@4	0.0 0	0.0 0	0.0 0	4.7 7	0.0 0	32.5 9	0.3 6	0.0 0	0.0 0	61.2 9	99.0 1
GTG-1-2-1L@5	0.0 0	0.0 0	0.0 8	3.3 3	0.0 0	32.6 9	0.3 0	0.0 0	0.0 0	62.6 9	99.0 9
GTG-1-2-1L@6	0.0 0	0.0 0	0.0 0	4.8 5	0.0 0	32.5 1	0.3 6	0.0 0	0.0 0	60.9 6	98.6 7
GJ-S1-5-2-1@1	0.0 0	0.0 0	0.0 0	4.6 9	0.0 0	32.3 2	0.3 0	0.0 0	0.0 0	61.6 6	98.9 7
GJ-S1-5-2-1@2	0.0 0	0.0 0	0.0 0	4.8 6	0.0 1	32.2 3	0.3 9	0.0 0	0.0 1	61.4 3	98.9 4
Sp3 n = 34											
GJ-S1-5-2-1@3	0.0 0	0.0 0	0.0 5	3.1 5	0.0 0	32.7 0	0.2 6	0.0 1	0.0 0	63.2 4	99.4 0

	0.0	0.0	0.0	3.2	0.0	32.3	0.3	0.0	0.0	62.5	98.5
GJ-S1-5-2-1@4	0	0	8	4	1	6	1	0	0	7	7
	0.0	0.0	0.0	3.3	0.0	32.4	0.2	0.0	0.0	62.5	98.5
GJ-S1-5-2-1@5	0	0	2	4	0	4	7	0	0	2	9
	0.0	0.0	0.0	3.4	0.0	32.7	0.2	0.0	0.0	62.4	98.9
GJ-S1-5-2-1@6	0	0	0	8	0	0	6	0	0	7	0
	0.0	0.0	0.0	3.4	0.0	32.6	0.3	0.0	0.0	62.2	98.8
GJG-1-36-1-SP@1	0	0	7	1	0	5	9	0	0	8	0
	0.0	0.0	0.0	3.4	0.0	32.3	0.3	0.0	0.0	62.6	98.6
GJG-1-36-1-SP@2	0	0	0	4	0	0	4	0	0	1	9
	0.0	0.0	0.0	3.2	0.0	32.4	0.3	0.0	0.0	62.5	98.6
GJG-1-36-1-SP@3	0	0	6	6	0	0	4	3	0	9	9
	0.0	0.0	0.0	3.5	0.0	32.4	0.3	0.0	0.0	63.0	99.4
GJG-1-36-1-SP@4	0	2	3	1	0	6	5	0	2	5	3
	0.0	0.0	0.0	3.4	0.0	32.5	0.3	0.0	0.0	62.7	99.0
GJG-1-36-1-SP@5	0	0	2	1	0	2	4	0	0	8	6
	0.0	0.0	0.1	3.5	0.0	32.0	0.3	0.0	0.0	62.5	98.6
GJG-1-36-1-SP@6	0	4	7	3	0	7	3	0	0	1	4
	0.0	0.0	0.0	2.3	0.0	32.5	0.2	0.0	0.0	63.7	99.0
GJ-N1-3-1@1	0	0	5	6	0	6	6	0	1	7	1
	0.0	0.0	0.0	2.3	0.0	32.6	0.2	0.0	0.0	64.0	99.2
GJ-N1-3-1@2	0	0	0	4	1	4	4	1	0	3	7
	0.0	0.0	0.0	2.3	0.0	32.4	0.2	0.0	0.0	63.7	98.8
GJ-N1-3-1@3	0	0	0	9	0	5	5	0	0	9	8

GJ-N1-3-1@4	0.0 0	0.0 0	0.0 6	2.4 9	0.0 0	32.4 7	0.2 4	0.0 0	0.0 0	63.8 7	99.1 2
GJ-N1-3-1@5	0.0 0	0.0 0	0.0 6	2.5 3	0.0 0	32.6 7	0.2 5	0.0 1	0.0 0	63.9 3	99.4 5
GJ-N1-3-1@6	0.0 0	0.0 0	0.0 7	2.5 1	0.0 0	32.3 4	0.2 6	0.0 0	0.0 1	63.8 8	99.0 7
GJ-S1-3B-1@1	0.0 0	0.0 0	0.0 0	5.0 5	0.0 1	32.3 2	0.3 2	0.0 1	0.0 0	61.0 2	98.7 2
GJ-S1-3B-1@2	0.0 0	0.0 0	0.0 0	4.2 1	0.0 1	32.0 9	0.2 6	0.0 0	0.0 1	62.0 2	98.6 0
GJ-S1-3B-1@3	0.0 0	0.0 0	0.0 1	5.0 9	0.0 0	32.4 2	0.3 6	0.0 0	0.0 0	60.9 6	98.8 3
GJ-S1-3B-1@4	0.0 0	0.0 0	0.0 0	3.1 7	0.0 1	32.6 9	0.2 6	0.0 1	0.0 1	62.6 8	98.8 2
GJ-S1-3B-1@5	0.0 0	0.0 0	0.0 0	4.9 3	0.0 0	32.0 7	0.2 9	0.0 1	0.0 0	61.3 0	98.6 0

(continued on next page)

Table 1. (continued)

Samples No.	Ge	Ga	Cu	Fe	Mn	S	Cd	In	Sn	Zn	Total
GJ-S1-3B-1@6	0.0 0	0.0 0	0.0 0	5.0 0	0.0 0	31.7 9	0.3 3	0.0 0	0.0 1	61.2 4	98.3 6

GJ-S1-3J-1@1	0.0 0	0.0 0	0.0 0	4.9 1	0.0 0	32.3 7	0.3 4	0.0 0	0.0 1	61.3 2	98.9 6
GJ-S1-3J-1@2	0.0 0	0.0 0	0.0 0	4.6 1	0.0 0	32.4 8	0.3 5	0.0 1	0.0 0	61.9 8	99.4 3
GJ-S1-3J-1@3	0.0 0	0.0 0	0.0 0	4.5 5	0.0 1	32.3 4	0.3 4	0.0 0	0.0 0	61.3 9	98.6 2
GJ-S1-3J-1@4	0.0 0	0.0 0	0.0 8	4.5 9	0.0 0	32.8 3	0.3 5	0.0 0	0.0 0	61.5 5	99.4 0
GJ-S1-3J-1@5	0.0 0	0.0 0	0.0 6	4.8 5	0.0 0	32.5 6	0.3 5	0.0 0	0.0 0	61.7 0	99.5 1
GJ-S1-3J-1@6	0.0 0	0.0 0	0.1 0	4.7 4	0.0 0	32.3 1	0.3 4	0.0 0	0.0 1	61.4 7	98.9 7
GTG-1-2-1S@1	0.0 0	0.0 0	0.0 0	3.6 5	0.0 0	32.5 9	0.3 3	0.0 0	0.0 0	62.5 0	99.0 6
GTG-1-2-1S@2	0.0 0	0.0 0	0.0 0	4.2 5	0.0 0	32.3 4	0.3 2	0.0 0	0.0 0	62.1 0	99.0 2
GTG-1-2-1S@3	0.0 0	0.0 0	0.0 0	4.1 4	0.0 0	32.3 5	0.3 0	0.0 0	0.0 0	62.0 7	98.8 6
GTG-1-2-1S@4	0.0 0	0.0 0	0.0 0	4.0 6	0.0 0	32.7 3	0.3 1	0.0 0	0.0 0	62.3 1	99.4 1
GTG-1-2-1S@5	0.0 0	0.0 0	0.0 5	4.8 2	0.0 0	32.4 9	0.3 7	0.0 0	0.0 0	60.9 4	98.6 8
GTG-1-2-1S@6	0.0 0	0.0 0	0.0 1	3.8 1	0.0 1	32.6 3	0.3 5	0.0 0	0.0 0	62.2 4	99.0 4

1019

1020

Appendix Table 2. Summary statistics of LA-ICP-MS data and temperature estimated results for sphalerite (ppm)

Samples	M														T(°C)
No.	S	n	Fe	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	Hg	Pb	)
Sp1 n = 25															
GB-1-1- SP@1	1470		182	20				10	385				56		20
	63.3	5.0	73.2	9.5	367	4.6	1.2	6.4	1.1	0.	22.	82.	4.3	125	6
	0	8	7	0	.68	9	6	5	1	16	96	80	9	.40	
GB-1-1- SP@2	1473		202	20					388	B			70		20
	48.8	6.3	05.5	7.4	31.	1.3	2.4	24.	4.5	D	1.5	0.4	5.2	30.	1
	9	0	4	9	58	6	0	24	1	L	3	3	0	33	
GB-1-1- SP@3	1478		202	20					388	B			63		20
	91.4	4.8	41.3	1.7	70.	0.9	2.3	40.	4.2	D	1.4	4.5	4.0	62.	3
	4	7	4	4	74	4	9	62	7	L	1	7	3	28	
GB-1-1- SP@4	1473		198	19					385	B			79		20
	84.7	4.2	12.9	3.7	49.	1.0	2.0	46.	6.7	D	1.3	6.2	9.6	62.	3
	6	7	7	4	32	5	5	50	7	L	9	3	3	95	
GB-1-1- SP@5	1490		198	20					376	B			66		21
	46.7	4.2	13.5	8.0	33.	0.5	1.3	54.	0.0	D	0.9	3.5	1.9	74.	4
	5	9	1	7	79	1	6	11	3	L	3	0	7	92	
GB-1-1- SP@6	1494		168	29				31	357			17	54		21
	81.2	10.	14.8	7.3	224	7.2	3.0	8.5	5.6	2.	81.	4.2	8.7	536	3
	9	86	0	0	.74	0	8	2	0	17	10	7	0	.75	

	1914		223						285	B			34		
GJG-1-	89.9	13.	71.9	88.	20.	15.	1.0	14.	8.5	D	0.7	0.3	9.7	6.0	19
13-1@1	7	39	8	05	06	81	5	04	7	L	3	6	9	9	3
	1814		207	10					254				31		19
GJG-1-	54.2	7.3	85.9	8.2	80.	8.4	1.1	44.	8.5	0.	14.	36.	0.4	10.	8
13-1@2	3	8	0	9	69	8	6	36	7	05	24	14	8	22	
	1848		218	11					261	B			31		18
GJG-1-	45.1	7.6	29.5	2.9	26.	21.	0.8	19.	7.9	D	0.8	1.5	0.4	10.	9
13-1@3	5	1	9	2	15	18	1	94	4	L	0	9	7	18	
	1860		219	11					263	B			31		19
GJG-1-	63.1	7.7	28.5	2.7	18.	7.8	1.3	24.	8.7	D	1.6	6.0	9.4	12.	3
13-1@4	9	2	9	3	37	0	0	31	2	L	1	8	4	79	
	1871		217	11					252				31		19
GJG-1-	26.3	7.6	99.9	2.0	15.	2.4	2.4	31.	9.6	0.	2.2	4.9	4.0	18.	5
13-1@5	3	3	9	1	52	2	0	62	3	01	1	4	9	06	
	1875		211	11					254				30		20
GJG-1-	33.2	7.5	59.1	9.9	10.	4.0	0.9	16.	0.4	0.	1.2	0.8	7.1	9.3	2
13-1@6	0	7	1	3	69	1	1	30	4	02	6	6	8	0	
	1826		105		433				381				45		18
GJ-N7-	42.9	4.9	03.5	23.	8.6	17.	2.5	42.	0.9	1.	12.	15.	0.2	119	2
22-1@1	7	0	6	58	2	01	3	05	8	40	33	18	7	.82	
	1829		105		472				380				39		18
GJ-N7-	35.4	3.1	56.9	31.	6.2	26.	0.8	52.	8.0	1.	158	82.	9.0	139	5
22-1@2	5	1	5	41	7	74	2	67	8	67	.27	59	2	.80	
	1832		126		495				384				35		23
GJ-N7-	45.7	4.8	37.6	46.	3.5	0.5	0.5	25.	8.4	1.	35.	7.8	9.2	68.	3
22-1@3	4	6	2	64	0	6	8	13	9	12	45	5	3	18	

	1849		106		767				361				52			
GJ-N7-22-1@4	50.7	BD	82.8	42.	8.7	26.	0.7	34.	6.4	1.	75.	24.	5.7	141	#	
	0	L	3	45	8	16	7	91	6	37	03	43	8	.60		
	1842		110		742				365				54			
GJ-N7-22-1@5	01.7	BD	82.0	34.	2.7	49.	0.8	59.	8.3	2.	275	68.	1.2	192	#	
	5	L	8	23	2	65	0	89	1	24	.93	69	2	.93		
	1908	31	127		432			12	407			22	42			
GJ-N7-22-1@6	73.2	7.8	83.0	32.	4.8	46.	0.9	6.7	9.1	2.	329	3.3	1.5	431	21	
	0	7	0	26	8	08	7	3	6	48	.88	1	2	.72	9	
GT-N7-18-Z-1@1	1994		217		103				316				38			20
	59.0	14.	03.8	64.	5.6	37.	0.5	29.	2.4	0.	19.	25.	1.2	60.	7	
	5	08	5	44	6	83	0	92	5	30	17	54	7	99		
GT-N7-18-Z-1@2	2021	47	292		334				306				31			23
	66.8	9.1	60.6	63.	3.2	23.	1.2	41.	9.4	0.	32.	25.	2.2	135	7	
	2	9	9	07	1	62	6	89	7	42	37	16	5	.64		
GT-N7-18-Z-1@3	1996		242		10				332				41			18
	58.8	15.	18.9	94.	848	8.5	2.7	34.	2.0	0.	5.6	9.6	0.0	61.	1	
	8	12	3	50	.51	3	3	61	9	30	5	0	4	33		

(continued on next page)

Table 2. (continued)

Samples No.	M														T(°C)
	S	n	Fe	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	Hg	Pb	
GT-N7-18-Z-1@4	2105	17	185	13	410			66	390			62	52		23
	48.4	1.1	61.2	0.8	4.8	10.	1.2	4.3	5.0	0.	183	1.3	3.6	228	7
	9	8	4	9	1	39	0	3	2	86	.84	8	9	.94	

GJ-1-3- 2@1	2084		264	13					353				41			19
	96.4	20.	68.9	1.2	253	44.	1.2	70.	0.0	0.	6.3	20.	1.5	73.		2
	2	91	4	2	.52	87	5	65	1	04	3	15	8	39		
GJ-1-3- 2@2	1997		226						330				39			21
	87.7	18.	78.5	96.	699	14.	0.9	27.	0.5	0.	4.1	3.2	7.7	54.		5
	5	18	4	48	.06	48	0	34	2	34	8	9	0	02		
GJ-1-3- 2@3	2012		183	14	375			14	292				46			22
	05.5	44.	70.6	3.4	5.2	18.	1.1	8.6	6.4	1.	89.	96.	8.3	868		4
	3	22	9	4	5	39	6	0	5	51	92	10	6	.21		
Sp2 n = 31																
GJ-1-3- 2@4	1828		337	10					347	B			74			19
	39.4	10.	32.0	5.9	151	37.	0.8	31.	2.6	D	1.6	15.	9.3	14.		2
	1	59	3	0	.23	68	0	46	8	L	1	08	6	12		
GJ-1-3- 2@5	1874		313						367	B			74			21
	50.3	13.	51.2	93.	73.	2.6	1.5	29.	6.2	D	1.0	19.	3.6	50.		3
	7	78	2	89	28	3	5	85	2	L	0	06	1	89		
GJ-1-3- 2@6	1963		318	10					388	B			65			18
	13.2	10.	73.9	8.7	47.	28.	2.2	36.	9.8	D	2.3	15.	4.9	25.		3
	4	71	5	0	08	59	5	69	8	L	9	08	2	29		
GT-N7- 18-Z- 1@5	2023		195	12	265				284				42			20
	78.2	16.	00.6	9.8	2.7	49.	1.0	84.	8.1	1.	46.	39.	9.2	59.		3
	6	19	4	5	0	01	3	48	4	08	13	86	3	87		
GT-N7- 18-Z- 1@6	2031		195	12					286				40			19
	83.2	11.	01.2	8.2	251	50.	0.9	29.	6.6	0.	2.0	5.4	9.8	20.		8
	0	54	9	5	.12	92	7	36	8	82	1	8	3	81		
GJG-1- 36J-1- SP@1	1547		218	35					374				43			20
	89.8	6.2	66.4	6.7	59.	16.	0.9	3.8	0.9	0.	1.1	1.2	5.2	31.		2
	5	2	5	8	66	39	4	7	4	29	2	7	5	74		

GJG-1-	1552		213	35					364				43			
36J-1-	27.5	6.0	54.8	8.4	145	12.	0.4	2.8	8.6	0.	9.8	6.2	9.4	22.		21
SP@2	8	5	2	3	.05	41	2	7	0	36	7	1	5	15		4
GJG-1-	1574		207	35					350				43			
36J-1-	42.6	5.9	51.4	7.7	358	9.4	1.0	4.3	5.3	0.	49.	17.	6.0	31.		20
SP@3	8	3	0	6	.76	8	5	5	3	53	58	52	4	82		9
GJG-1-	1927		223	32					366				51			
36J-1-	38.7	5.3	78.2	9.7	42.	16.	1.7	12.	3.1	0.	2.0	3.4	4.9	8.2		19
SP@4	2	6	9	9	16	81	8	69	9	27	4	5	9	2		3
GJG-1-	1928		224	33					372				50			
36J-1-	42.3	5.4	09.9	6.8	47.	15.	1.6	12.	7.1	0.	7.5	0.8	2.7	26.		19
SP@5	7	7	6	7	06	10	3	47	5	35	4	4	7	83		7
GJG-1-	1965		225	33					368				50			
36J-1-	00.3	5.4	48.3	6.4	48.	17.	1.8	14.	5.8	0.	5.7	2.7	3.7	16.		19
SP@6	5	5	4	4	28	01	8	50	5	47	4	3	3	59		5
GJG-1-5-	1513		219	32					350				47			
1-SP@1	89.2	44.	77.4	2.7	98.	67.	1.5	9.1	9.6	1.	11.	11.	7.0	17.		20
	0	49	7	4	88	08	9	9	1	30	88	29	2	42		8
GJG-1-5-	1528		220	33					371				48			
1-SP@2	53.7	5.7	62.0	9.4	278	38.	6.3	6.9	7.5	1.	194	1.1	1.9	5.1		18
	4	4	1	4	.03	91	1	3	5	67	.82	9	3	4		2
GJG-1-5-	1515		218	34					372				49			
1-SP@3	37.9	5.6	82.1	0.7	215	72.	16.	10.	1.0	1.	85.	6.5	3.1	7.3		16
	9	2	6	6	.66	39	21	02	7	08	37	1	8	2		4
GJG-1-5-	1538		226	35					392				49			
1-SP@4	84.2	5.8	35.7	7.0	435	40.	9.0	8.6	1.3	2.	329	1.7	6.9	5.5		18
	3	1	1	8	.72	46	2	7	6	12	.35	1	7	1		0

	1562		232	36					396				55			
GJG-1-5-	86.6	6.6	11.5	2.8	509	44.	3.9	9.4	3.6	2.	400	4.4	9.5	14.		18
1-SP@5	3	8	5	8	.02	97	6	8	3	45	.18	0	1	06		9
	1574		234	36					381			178		52		
GJG-1-5-	58.9	20.	51.2	0.8	203	53.	7.2	10.	0.0	8.	7.5	5.8	2.7	32.		19
1-SP@6	6	08	1	6	.56	29	0	88	1	36	3	8	8	28		9

(continued on next page)

Table 2. (continued)

Samples	M														T(
No.	S	n	Fe	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	Hg	Pb	°C
	1991		151	12	135				305				54	108	
GJ-S1-	48.2	13.	79.0	2.5	5.9	7.1	1.1	88.	5.9	1.	58.	58.	5.3	1.4	21
11-1@1	6	48	9	0	8	9	2	24	6	57	71	57	1	3	9
	1993		194	12	134				287				45		
GJ-S1-	21.7	10.	00.4	7.9	1.2	13.	1.3	55.	2.7	1.	5.7	23.	7.4	60.	20
11-1@2	5	84	4	9	6	67	9	95	4	26	1	45	2	12	9
	2053		282	10					308	B			44		
GJ-S1-	89.5	21.	08.7	6.3	184	30.	1.1	41.	5.0	D	2.4	20.	0.3	19.	19
11-1@3	3	26	1	8	.32	05	9	20	3	L	3	48	0	55	5
	1908		213						300				38		
GJ-S1-	17.4	12.	99.2	86.	45.	4.9	1.0	14.	0.2	0.	0.7	0.0	9.4	5.9	20
11-1@4	2	40	5	57	24	5	0	47	3	02	5	0	9	4	5
	1976		311	10					376	B			65		
GJ-S1-	67.4	11.	00.9	5.9	149	79.	1.0	30.	8.7	D	1.3	5.8	9.2	30.	18
11-1@5	5	15	8	6	.84	97	2	48	5	L	0	1	1	42	2

	1978		295	10					368				67			
GJ-S1-	29.0	9.8	54.8	1.7	631	37.	2.5	42.	9.5	0.	3.4	24.	0.2	46.		18
11-1@6	0	5	5	5	.59	73	2	69	5	05	3	74	7	76		4
	1930		220						286	B			35			
GTG-1-2-	65.9	13.	27.2	90.	182	1.9	1.0	25.	8.7	D	1.5	3.0	5.8	12.		21
1L@1	4	43	3	13	.18	2	8	35	8	L	9	2	9	04		4
	2038		295	12					383				42			
GTG-1-2-	23.1	14.	47.7	2.9	858	5.4	0.8	27.	5.4	0.	2.9	3.4	2.8	24.		21
1L@2	6	41	5	9	.16	6	6	61	3	03	7	2	6	85		5
	2056		328	12					385	B			42			
GTG-1-2-	13.9	16.	03.6	7.3	645	4.6	0.7	15.	9.4	D	2.1	2.7	8.6	8.9		21
1L@3	1	01	5	4	.22	9	5	90	4	L	4	6	6	8		7
	2048		322	12	132				369				36			
GTG-1-2-	62.6	15.	03.6	7.8	7.4	37.	0.8	52.	6.0	0.	3.1	14.	6.4	27.		19
1L@4	0	60	7	7	3	67	8	61	9	02	2	52	3	32		5
	1977	31	291		251				309				37			
GTG-1-2-	49.2	8.0	33.6	71.	2.8	67.	65.	50.	6.9	0.	140	29.	7.9	111		18
1L@5	7	8	7	73	8	79	81	14	4	85	.44	66	8	.81		5
	2056		327	12					373				37			
GTG-1-2-	46.8	17.	14.5	7.3	669	35.	1.7	36.	2.6	0.	3.4	10.	0.0	22.		19
1L@6	4	75	0	9	.30	41	8	04	9	04	5	01	2	31		3
	1963		311						378				64			
GJ-S1-5-	42.9	10.	21.8	99.	73.	60.	1.9	37.	3.8	0.	1.5	8.3	1.7	27.		17
2-1@1	1	47	2	60	23	21	8	60	6	02	0	7	0	68		8
	1962		310	10					388				65			
GJ-S1-5-	89.9	10.	38.1	5.9	26.	6.5	1.2	23.	6.8	0.	2.1	10.	5.4	20.		21
2-1@2	6	27	3	8	14	0	5	48	4	04	5	38	4	18		0

	2029		211	12	100	10			289				41		
GJ-S1-5-2-1@3	80.9	15.	35.2	9.7	3.4	1.5	29.	66.	9.2	0.	12.	26.	4.2	44.	160
	1	79	3	5	1	8	03	36	8	87	54	57	9	90	
	1903		209			36	14		287				36		118
GJ-S1-5-2-1@4	25.8	12.	18.1	84.	525	4.0	1.3	74.	3.1	0.	14.	25.	8.9	10.	
	5	89	7	02	.32	1	1	62	9	09	81	34	0	53	
	1911		220			20			289	B			36		133
GJ-S1-5-2-1@5	60.5	14.	63.6	86.	315	5.4	25.	57.	4.9	D	2.0	7.3	3.2	20.	
	8	07	1	75	.43	1	34	79	5	L	3	9	9	95	
	1915		221			18	20		281	B			35		113
GJ-S1-5-2-1@6	25.9	14.	37.9	86.	182	7.8	4.1	36.	3.7	D	1.3	6.6	3.5	8.7	
	0	13	6	59	.31	2	4	20	8	L	3	5	0	4	
GJG-1-36-1-SP@1	1542		221	37		51			390				43		123
	80.9	6.3	96.3	7.4	733	4.1	78.	7.2	2.3	0.	91.	6.1	3.2	6.4	
	2	8	4	4	.28	4	27	7	2	47	56	1	5	4	
GJG-1-36-1-SP@2	1551		218	36		34			370				43		133
	09.3	6.0	87.8	7.4	629	6.5	54.	6.0	0.8	0.	161	7.1	6.7	12.	
	8	5	8	0	.83	6	03	7	7	76	.65	7	0	72	
GJG-1-36-1-SP@3	1323		180	29					304				37		156
	74.1	4.9	70.9	9.6	272	57.	14.	3.3	1.8	0.	16.	6.1	9.5	24.	
	3	5	6	2	.75	50	80	0	9	25	43	2	1	85	

(continued on next page)

Table 2. (continued)

Samples	Mn	Pb	Cd	Hg	Co	Cu	Ga	Ge	Ag	In	Sn	Sb	Tl	T(°C)	
No.	S	n	Fe	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	Hg	Pb	)

GJG-1-	1927		224	33		46	13		366				52			
36-1-	55.7	5.4	40.8	4.7	742	4.7	2.7	29.	6.6	0.	42.	7.7	1.0	13.		11
SP@4	5	7	2	3	.05	4	8	90	8	38	63	4	8	89		7
GJG-1-	1998		227	34		29			363				53			
36-1-	19.2	4.8	46.9	3.0	369	4.0	29.	25.	9.4	0.	9.7	6.0	1.5	19.		13
SP@5	5	0	0	3	.97	8	60	91	1	15	5	2	0	39		0
GJG-1-	1996		230	36		34	10		350				52			
36-1-	06.9	5.8	15.1	9.0	561	9.2	8.3	24.	9.9	0.	27.	5.9	8.5	10.		12
SP@6	8	1	0	2	.15	0	6	07	2	25	15	7	6	24		0
	2019		155			10	41		252				49			
GJ-N1-3-	14.6	3.6	39.5	66.	779	8.8	7.4	88.	2.7	0.	21.	17.	9.8	23.		10
1@1	1	3	9	59	.96	2	5	08	0	09	47	08	2	81		2
	2025		157			15			257				52			
GJ-N1-3-	74.9	4.5	42.7	78.	110	87.	6.1	65.	6.7	0.	7.7	28.	4.1	34.		11
1@2	6	4	2	50	.37	71	4	63	0	04	8	38	9	94		2
	2021		164			34			256	B			51			
GJ-N1-3-	82.9	8.3	57.8	77.	83.	74.	7.8	64.	8.0	D	2.0	11.	6.5	19.		10
1@3	6	9	2	75	06	03	6	79	6	L	7	15	4	88		7
	2043		168			47			262				50			
GJ-N1-3-	13.7	6.1	77.7	84.	728	73.	2.8	73.	5.5	0.	11.	40.	9.8	10.		10
1@4	8	7	1	60	.39	04	3	63	0	06	24	10	0	17		8
	2042		170			33			267				47			
GJ-N1-3-	94.7	7.9	48.6	86.	595	52.	1.0	62.	6.7	0.	4.5	10.	1.3	10.		11
1@5	9	2	6	86	.91	65	8	11	4	04	9	52	4	32		5
	2049		170			44	10		264				48			
GJ-N1-3-	80.3	8.6	11.2	86.	769	38.	3.8	3.4	5.3	0.	12.	60.	2.4	22.		11
1@6	5	7	1	20	.06	99	1	3	0	06	54	48	7	10		9

	1834		330				27		349	B			77			
GJ-S1-	07.0	11.	38.9	94.	356	52.	0.0	27.	8.7	D	6.4	8.7	9.3	12.		12
3B-1@1	9	01	3	33	.61	92	0	04	0	L	9	5	1	82		8
	1850		339				22		341				74			
GJ-S1-	35.9	27.	90.7	98.	518	62.	0.3	50.	5.2	0.	21.	27.	4.6	17.		14
3B-1@2	6	64	8	46	.55	11	7	18	1	08	32	64	9	16		3
	1822		335	10			26	11		347	B		76			
GJ-S1-	08.3	10.	78.0	3.9	295	4.1	0.9	37.	1.4	D	2.2	3.6	2.6	4.3		12
3B-1@3	9	58	6	7	.52	3	4	36	0	L	8	1	2	1		1
	1834		217	11			16		254	B			32			
GJ-S1-	36.1	7.8	71.8	5.9	69.	24.	6.7	45.	4.9	D	2.2	9.0	0.9	25.		12
3B-1@4	3	2	4	2	32	23	4	01	3	L	2	7	7	71		9
	1868		330	10					354	B			71			
GJ-S1-	17.0	10.	25.9	4.4	167	90.	1.3	35.	5.6	D	1.3	16.	7.1	20.		17
3B-1@5	7	41	8	6	.24	22	9	96	3	L	1	34	2	09		7
	1854		333				10		338				65			
GJ-S1-	05.5	15.	79.6	97.	737	27.	9.3	64.	1.2	0.	12.	37.	5.0	46.		15
3B-1@6	1	03	7	35	.88	12	2	36	4	05	06	01	7	14		3
	1958		317	10			14		389	B			68			
GJ-S1-3J-	07.1	10.	99.2	6.3	264	7.9	11.	43.	1.5	D	1.9	14.	6.9	28.		14
1@1	7	99	3	2	.58	4	28	52	5	L	6	39	7	24		9
	1875		312						371	B			68			
GJ-S1-3J-	99.0	13.	17.9	99.	424	38.	17.	37.	2.8	D	1.3	17.	3.7	737		16
1@2	9	35	9	38	.65	23	72	32	3	L	3	12	4	.00		0
	1866		296				33		359	B			67			
GJ-S1-3J-	95.9	12.	67.0	97.	759	38.	0.5	45.	7.9	D	1.2	26.	7.1	47.		12
1@3	1	85	2	25	.51	11	7	68	2	L	7	46	2	23		8

	1874		303	10	243				373			74				
GJ-S1-3J-1@4	44.3	14.	02.9	9.0	3.3	36.	2.9	55.	0.0	0.	1.6	16.	7.3	58.	17	
	9	17	7	7	9	56	8	05	5	01	6	38	6	69	9	
	1898		328	12				28		368			70			
GJ-S1-3J-1@5	38.9	15.	04.8	1.9	684	77.	0.2	48.	8.5	0.	8.4	35.	7.3	14.	13	
	0	84	1	6	.09	79	5	21	9	03	9	89	7	32	0	
	1899		297	12	114				352	B			56			
GJ-S1-3J-1@6	26.6	14.	05.9	3.9	6.1	21.	5.3	53.	5.2	D	5.7	39.	5.1	82.	18	
	1	74	2	3	8	68	9	60	6	L	3	85	9	74	0	
	1988		236						328				35			
GTG-1-2-1S@1	08.0	15.	92.0	76.	645	77.	63.	28.	0.7	0.	4.6	10.	9.8	52.	15	
	8	06	1	87	.61	10	46	16	6	23	9	15	5	78	0	

(continued on next page)

Table 2. (continued)

Samples	M															T(°C)
No.	S	n	Fe	Co	Cu	Ga	Ge	Ag	Cd	In	Sn	Sb	Hg	Pb		
	2069		294	12				62		321	B		44			
GTG-1-2-1S@2	17.0	21.	08.6	3.9	42.	34.	1.5	28.	9.2	D	2.4	8.3	6.2	18.	12	
	7	48	0	1	79	79	4	47	3	L	8	1	4	31	7	
	2092		280	12				11		335	B		46			
GTG-1-2-1S@3	36.6	21.	20.2	5.1	235	86.	8.4	36.	3.7	D	2.9	12.	4.1	15.	13	
	8	37	3	4	.17	25	7	12	0	L	8	18	9	58	4	
	2096		271	13					349	B			44			
GTG-1-2-1S@4	76.5	21.	37.9	0.0	259	75.	25.	51.	6.7	D	5.0	20.	8.3	28.	15	
	3	86	6	9	.88	96	22	92	5	L	4	52	5	08	3	

	2048		315	12	279	23	10	12	371				38		
GTG-1-2-	77.0	15.	72.3	6.7	1.9	3.9	7.0	3.4	4.6	0.	130	77.	3.8	40.	14
1S@5	0	44	5	4	0	5	7	4	2	57	.59	47	9	17	3
	2095		258	13					368				41		
GTG-1-2-	32.9	19.	72.0	3.5	363	22.	22.	33.	6.8	0.	3.4	10.	9.9	143	16
1S@6	2	21	8	8	.18	75	45	25	2	04	2	29	3	.39	8

BDL means concentration low than detection limits

1021

1022